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I. Introduction

Sky Solutions (Sky) services help federal agencies accelerate transformation through artificial intelligence (AI), digital modernization, and cybersecurity. We offer advanced analytics, low-code/no-code development, DevSecOps, cloud migration, and zero-trust security solutions. Sky modernizes legacy systems using a human-centered, agile approach, enabling secure, data-driven decision-making. Our team supports key sectors, including healthcare, financial services, and national security, with clients including CMS, IRS, DOJ, and FEMA. Sky delivers scalable, efficient, compliant solutions tailored to mission needs.

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II. Responses to Questions

RFI Section B. Patients and Caregivers

1. Patient Needs

***PC-1.* What health management or care navigation apps would help you understand and manage your (or your loved ones') health needs, as well as the actions you should take?**

a. What are the top things you would like to be able to do for you or your loved ones' health that can be enabled by digital health products?

A digital health product that unifies **data centralization**, **proactive guidance**, **humanized education**, and **family support** has the power to transform healthcare from a fragmented, reactive experience into a continuous, connected journey. By consolidating medical records, data from wearable devices, prescriptions, and care histories in one place, users can gain a holistic view of their health without the stress of navigating multiple platforms or repeating information. Proactive guidance, powered by AI, enables timely, personalized recommendations, helping users know when to seek care, adjust lifestyle choices, or manage medications. Patient-focused education ensures that health information is not only accurate but also accessible, using clear language, visuals, and context-specific learning to empower decision-making. For caregivers and families, integrated support tools allow collaborative care management—tracking appointments, medications, and conditions across multiple profiles. Whether you're managing a chronic condition, recovering post-surgery, or supporting an aging parent, this type of digital health solution becomes more than just a tracker; it's a reliable companion that simplifies complex terminology, anticipates needs, and supports better health outcomes at every step of the patient's healthcare journey. Examples of such digital health products include:

1. Centralized Medical Records Access

Imagine having one app that stores all your medical data securely, including lab results, prescriptions, imaging reports, past diagnoses, vaccination records, and visit summaries. This feature should:

- Allow uploads from clinics, hospitals, and labs.
- Enable easy patient information sharing with healthcare providers during appointments or emergencies.

- Support multi-language and user-friendly formats.
- Include alerts for upcoming follow-ups based on previous visits.

2. Personalized Health Dashboards
Health data should be easy to understand. Dashboards could display:
- Real-time vitals from wearable devices, including heart rate, oxygen saturation, sleep patterns, or glucose levels
- Graphs showing changes in metrics over time
- Upcoming appointments and medication schedules
- Custom health indicators for chronic conditions or specific goals (e.g., pregnancy, recovery, training)

3. Symptom Checker and Action Guidance
AI-based symptom checkers could:
- Ask intelligent questions to assess the severity
- Offer initial guidance (e.g., home care vs. telehealth vs. urgent care)
- Provide clear visual risk indicators (red/yellow/green)
- Integrate local provider directories and emergency contact details based on recommendations

4 Care Navigation Support
It can be overwhelming to navigate healthcare systems. The app should:
- Match symptoms or conditions with the correct specialist
- Help locate nearby in-network doctors and available appointments
- Offer step-by-step pathways for procedures like surgeries or childbirth
- Explain insurance authorizations and next steps in plain language

5 Medication Management
To reduce errors and improve adherence, the app could:
- Send automatic reminders for each medication
- Track doses taken or missed with alerts
- Flag potential interactions based on your medication list
- Sync prescriptions with local pharmacies for easy refills

6 Chronic Disease Management Tools
Specialized features can make long-term condition management easier:
- Tracking blood sugar levels for diabetics, BP for hypertensive patients, or peak flow for asthma patients
- Provide real-time feedback or alerts when readings fall outside the safe range
- Offer evidence-based lifestyle tips and routine checklists
- Connect patients with health coaches or care coordinators

7. Health Literacy and Decision Support
Most patients feel overwhelmed by medical jargon. The app should:
- Break down test results into simple terms (e.g., what does “low hemoglobin” mean?)
- Offer explanations via text, video, or voice assistant

- Help you understand treatment options with pros, cons, and expected outcomes
- Include decision trees or quizzes to clarify what action to take next

8 Caregiver Mode / Family Health Tracker
Managing a loved one's care (e.g., elderly parent, child) should be simple:
- Switch between profiles for different family members
- Receive alerts for their medications, appointments, and vitals
- Share access securely with other family members or legal guardians
- Include tools for caregiver stress, respite planning, and progress notes

9 Mental Health and Wellness Features
Mental and emotional health must be part of the system, so an ideal app would:
- Include mood journaling, guided meditation, and breathing exercises
- Offer access to licensed therapists through chat or video
- Monitor behavioral trends like sleep quality, screen time, or isolation
- Provide mental health screenings (e.g., for anxiety or depression) and follow-up support

10. Preventive Health and Goal Tracking
Prevention is better than cure. The app should:
- Track when vaccines, mammograms, colonoscopies, or dental check-ups are due
- Suggest goals such as walking 10,000 steps/day, drinking more water, or improving sleep
- Offer motivational tips, reminders, or badges to keep patients engaged
- Sync with fitness devices and allow goal-based coaching or challenges

b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.

If I had a personal assistant to support my health needs, key functions things I would ask them to help with, many of which could be efficiently supported by software solutions in the future:

1. **Provide Personalized Health Recommendations:** Based on my data, I'd want them to suggest dietary tips, fitness routines, or stress-reduction techniques relevant to my condition or goals.
2. **Push Me to Stay Active:** I'd want my assistant to act as a motivator, nudging me to work out when I'm skipping sessions, recommending quick routines on busy days, or gamifying fitness goals to keep me engaged and consistent.
3. **Suggest Meal Plans Based on My BMI and Health Conditions:** I'd want the assistant to offer tailored meal plans that align with my BMI, allergies, dietary preferences, and health goals, whether it's weight loss, managing blood sugar, or improving heart health. It could even integrate with grocery apps and suggest recipes based on what I have at home.
4. **Recommend the Best Doctors Nearby:** Based on my symptoms or ongoing needs, I'd like it to recommend top-rated doctors in my area, with consideration to specialization, availability, insurance coverage, patient reviews, and even gender, cultural, or language preferences.

5. **Manage Appointments and Scheduling:** I'd ask a smart assistant to find the best times for doctor visits, book appointments, send reminders, and reschedule if needed. They could also sync calendars, suggest nearby providers, and handle waitlists automatically.
6. **Track Medications and Refills:** I'd rely on the assistant to remind me to take medications on time, alert me about potential drug interactions, automatically request refills or check insurance coverage.
7. **Interpret Test Results and Reports:** I'd want help translating complex medical results into clear, actionable insights and advice, such as which results are normal, which are not, and what are the optimal next steps.
8. **Monitor Symptoms and Trends:** I'd expect the assistant to keep a log of symptoms, mood, energy levels, or vitals, and highlight patterns that might require medical attention.
9. **Support Caregiving Roles:** For loved ones, the smart assistant should help track their health status, appointments, medications, and share updates securely across authorized family members or caregivers.
10. **Navigate Insurance and Costs:** I'd want it to simplify insurance claims, track deductibles, explain coverage options, and estimate out-of-pocket expenses.

***PC-2.* Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?**

a. If so, what are some examples of benefits it has provided?

Yes, I currently use a centralized digital health platform that consolidates medical information for both me and my loved ones. This system acts as a unified patient portal, bringing together a wide range of healthcare data from medical histories and prescriptions to test results, vaccination records, and appointment schedules all in one secure and accessible location. Having this information centralized has significantly improved how my family manages our health, especially when coordinating care for elderly family members or individuals with chronic conditions. Instead of juggling multiple systems, doctor offices, or paperwork, I can quickly retrieve and share necessary information with healthcare providers, reducing delays in treatment and minimizing the risk of medical errors. The platform also offers features like medication reminders, lab result notifications, and appointment tracking, which help us stay abreast of ongoing care tasks. In addition, this platform empowers us to be more proactive by allowing real-time access to health updates and making it easier to prepare for consultations and monitor treatment progress. Ultimately, this level of integration not only improves communication and efficiency but also brings peace of mind, knowing that our health records are complete, up to date, and ready whenever they are needed.

Examples of the Benefits Realized:

1. **Improved Care Coordination**
   The centralized portal enables seamless communication across healthcare providers by making complete medical histories, test results, and prescriptions accessible in one location. This reduces duplication of tests and ensures consistent, informed decision-making.
2. **Enhanced Emergency Preparedness**
   In critical situations, having instant access to allergy information, existing conditions, and prior treatments for myself and my dependents has been crucial for prompt, accurate care.

3. **Efficient Medication and Appointment Management**
   The platform includes features like automated prescription refill alerts, dosage instructions, and appointment scheduling tools, minimizing the risk of missed doses or follow-ups, particularly critical needs in managing care for elderly family members.
4. **Data Transparency and Patient Empowerment**
   Easy access to health records promotes patient engagement, allowing us to review and better understand diagnoses, treatment options, and progress over time. It also supports informed decision-making and fosters a sense of control over personal and family health journeys.
5. **Time and Cost Savings**
   Consolidating all relevant health data into a single system has reduced administrative overhead, saved time in accessing information, and improved the overall efficiency of healthcare interactions.

To summarize, centralized access to health data not only streamlines care delivery but significantly enhances the quality of healthcare management for both individuals and families.

b. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?

c. Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?

Yes, while the centralized health platform I use provides access to a broad range of data, there have been noticeable gaps in the availability and usability of certain critical information. Imaging results such as X-rays, MRIs, lab results and CT scans are often not fully accessible within the portal—either limited to summary notes or lacking actual image files. This restricts the ability to share complete diagnostic information with specialists or obtain second opinions.

Additionally, the inability to retrieve older reports such as previous lab results, surgical notes, or treatment summaries can be challenging. Many portals only store recent data or require manual requests for archived records, which delays care planning and makes it harder to track long-term health trends and therefore manage chronic conditions optimally.

Other key obstacles include:

- **Fragmentation Across Systems**:
  Healthcare data is often distributed across multiple disconnected systems used by different hospitals, clinics, and specialists. For example, a specialist's notes or imaging results might not be available in the primary care provider's portal. This lack of interoperability makes it difficult to build a complete and continuous health profile, especially when transitioning between providers or facilities.

- **Limited Functionality and User Experience**:
  Many digital health platforms lack intuitive design or robust features. Test results may be displayed without reference ranges, historical comparisons, or plain-language

explanations, making them difficult for patients to interpret without medical help. The absence of search tools, data filters, or visualization options further limits usability, especially for managing chronic conditions or understanding trends over time.

- **Challenges in Referencing Historical Reports**:
  Accessing older medical reports such as labs from years prior, past surgical summaries, or long-term treatment histories is often difficult. Some systems only show recent data or require manual requests for archived information. This limits a patient's or caregiver's ability to track longitudinal health trends or provide comprehensive history during new consultations.

- **Technology Access and Digital Literacy**:
  Not all patients or family members have reliable access to the internet, smartphones, or computers. Additionally, older adults or those with low digital literacy often struggle to navigate health portals. Without support, they may miss critical updates, appointment reminders, or medication instructions, which can negatively impact their health outcomes.

These gaps reduce the full potential of digital health systems in supporting informed, continuous, and coordinated care.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

Based on our expertise and market research, we are aware of several health management, care navigation, and personal health record (PHR) applications that offer significant potential to support Medicare beneficiaries and their caregivers.

These apps typically focus on features such as medication management, appointment scheduling, chronic disease tracking, secure sharing of health information with caregivers and providers, and educational resources tailored to senior health needs. Examples of widely used platforms include:

- **MyChart**: Offers integration with healthcare providers' electronic health records (EHR) systems, allowing beneficiaries to view medical records, schedule appointments, and communicate with providers.
- **Medisafe**: Provides medication management with reminders, tracking, and drug interaction warnings to improve adherence.
- **CareZone**: Facilitates medication tracking, health data organization, and secure sharing of information with caregivers.
- **HealthVault (Microsoft)**: Although discontinued, it pioneered personal health records and inspired several current platforms focusing on data centralization and sharing.
- **Apple Health**: Aggregates health data from various sources and devices, enabling users to monitor vital signs, activity, and share data with providers.

Emerging AI-powered tools are also enhancing capabilities with features such as personalized care navigation, predictive health alerts, and tailored health recommendations. Notable examples include:

- **Ginger.io**: Uses AI to provide mental health coaching and support by monitoring mood and behavior patterns for timely intervention.

- **Ada Health**: Employs AI-driven symptom assessment to guide users toward appropriate care pathways.
- **Sensely**: Combines AI chatbots and avatar technology to offer virtual health coaching and navigation assistance tailored to individual needs.
- **Buoy Health**: An AI-powered symptom checker that helps users' self-triage and find appropriate care options.
- **Wellthie**: Provides AI-based care coordination and patient engagement tools, focusing on chronic disease management and social determinants of health.
- **CarePredict**: Utilizes AI and wearable sensors to monitor seniors' daily activities and detect early signs of health decline, enabling proactive interventions.

While these technologies show promise, adoption among Medicare beneficiaries may be constrained by factors such as digital literacy, access to devices, and user experience (UX) design. Hence, future solutions must prioritize simplicity, accessibility, and robust support for caregivers.

We advocate for continued collaboration between technology developers, healthcare providers, and Medicare stakeholders to refine and optimize these tools to meet the unique needs of Medicare beneficiaries and their caregivers.

PC-4. What features are missing from apps you use or that you are aware of today?

While many existing health management and care navigation apps offer valuable functionalities, there remain several key features that are either missing or underdeveloped, limiting their effectiveness for users, especially Medicare beneficiaries and their caregivers:

1. **Comprehensive Data Integration Across Providers** Most apps lack seamless interoperability with diverse healthcare systems, leading to fragmented health data. Users often cannot access a unified view that includes records from all providers, labs, and specialists.
2. **Historical Data Accessibility and Longitudinal Tracking** Many platforms do not adequately support easy access to older medical reports or the ability to track health trends over extended periods. This hampers users' ability to manage chronic conditions and make informed decisions based on historical health patterns.
3. **Personalized, Actionable Guidance** While some apps offer general health tips, few provide truly personalized recommendations or proactive alerts tailored to an individual's unique medical history, lifestyle, or risk factors.
4. **Robust Caregiver Collaboration Tools** There is often limited functionality that would enable caregivers to easily access, coordinate, and communicate about a loved one's health data, appointments, and medication management.
5. **User-Friendly Design for Older Adults** Many apps are not optimized for seniors or users with limited digital literacy, featuring complex navigation or unclear language, which reduces adoption and sustained use.
6. **Holistic Health Tracking** Integration of mental health, nutrition, physical activity, and social determinants of health within a single platform remains scarce yet is critical for comprehensive care management.

7. **Offline Access and Low Bandwidth Functionality** Inconsistent internet access or low bandwidth environments significantly impact usability, but few apps provide offline capabilities or simplified data syncing.

Addressing these gaps will be crucial to developing digital health solutions that truly empower Medicare beneficiaries and their caregivers to manage health effectively and improve outcomes.

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

Despite the rapid growth of digital health technologies, several types of apps that could profoundly enhance health management and care navigation for Medicare beneficiaries and their caregivers are still absent or underdeveloped.

Examples:

1. **Fully Integrated, Family-Centric Health Management Platforms.** An app that centralizes the health data of an entire family or care network, combining medical records, medication schedules, appointments, and real-time health monitoring, while allowing tailored access and communication among all members. Existing apps tend to focus on individuals rather than caregiving units.
2. **AI-Powered Predictive Care Navigation Assistants.** Intelligent virtual assistants that proactively analyze a user's health data, lifestyle, and social factors to predict risks and recommend personalized interventions or escalate care before acute issues arise. While some symptom checkers exist, fully predictive, longitudinal care navigation remains nascent.
3. **Cross-Provider, Interoperable Health Record Aggregators.** Platforms that can seamlessly pull data from any healthcare provider's system, regardless of technology or vendor, creating a truly unified patient record that's continuously updated and accessible to users and their care teams.
4. **Context-Aware Mental Health and Social Support Apps.** Tools that integrate mental health monitoring with social determinants of health, providing context-aware interventions, community resource recommendations, and real-time crisis support tailored to Medicare beneficiaries' unique circumstances.
5. **Offline-First, Low-Tech Digital Health Solution.** Apps designed specifically for populations with limited internet access or low digital literacy that can operate offline or with minimal connectivity, ensuring equitable access to digital health tools.

Factors that have hampered the development of such apps include:

- **Complexity of Healthcare Data Interoperability:** The healthcare ecosystem is fragmented with multiple EHR vendors, regulatory constraints, and data privacy concerns, making seamless data integration a major challenge.
- **Limited Focus on Care Networks:** Most digital health solutions remain patient-centric rather than caregiver- or family-centric, partly due to complexities in managing permissions, privacy, and varied user needs.
- **Technological and AI Maturity:** While AI shows promise, developing truly predictive, personalized care navigation systems requires vast, high-quality data sets, advanced algorithms, and rigorous clinical validation, which are still evolving.

- **Digital Divide and Accessibility Challenges:** Creating sophisticated yet user-friendly apps that serve older adults or low-resource populations demands significant design and development efforts, which may be underfunded or deprioritized.
- **Regulatory and Reimbursement Barriers:** Health apps that integrate deeply with clinical care often face complex regulatory pathways and uncertain reimbursement models, slowing innovation and adoption.

Addressing these challenges through cross-sector collaboration, user-centered design, and policy innovation is essential to unlocking the next generation of digital health tools for Medicare beneficiaries and beyond.

b. What set of workflows do you believe CMS is uniquely positioned to offer?

The Centers for Medicare & Medicaid Services (CMS) occupies a pivotal role in the U.S. healthcare ecosystem and is uniquely positioned to implement and scale workflows that can improve health outcomes for Medicare beneficiaries. Given its extensive regulatory authority, data access, and partnerships with healthcare providers, CMS can offer the following distinctive workflows:

1. **Centralized Patient Data Aggregation and Interoperability.** CMS can lead in creating standardized workflows that enable secure aggregation and seamless exchange of health information across diverse providers and health IT systems. This would facilitate a unified patient view, improve care coordination and reduce duplicative services.
2. **Beneficiary-Centric Care Coordination.** Leveraging its data and provider network, CMS can implement workflows that proactively identify high-risk beneficiaries and trigger coordinated care plans involving multidisciplinary teams, including primary care, specialists, social services, and caregivers.
3. **Outcome-Based Quality Measurement and Reporting.** CMS is uniquely positioned to develop and mandate workflows that collect, analyze, and report clinical and patient-reported outcomes, directly linking care quality to reimbursement models. This would drive provider accountability and continuous improvement.
4. **Integrated Medication Management and Adherence Support.** CMS can establish workflows that monitor prescription fills, flag potential drug interactions, and support medication adherence through partnerships with pharmacies, providers, and digital health tools.
5. **Beneficiary Education and Engagement.** With broad reach, CMS can design workflows that deliver personalized education, reminders, and decision support tools to beneficiaries, helping them better understand their conditions and treatment options.
6. **Claims-Driven Predictive Analytics and Risk Stratification.** Utilizing vast claims data, CMS can implement workflows that identify emerging health risks and social determinants impacting beneficiaries, enabling timely interventions before conditions worsen.
7. **Caregiver Support and Authorization Management.** CMS can pioneer workflows that formally integrate caregivers into the care process, managing permissions and facilitating communication to support holistic beneficiary care.

By leveraging its regulatory authority, data resources, and provider relationships, CMS is uniquely positioned to develop these workflows at scale, thereby enhancing care quality, reducing costs, and improving the health and wellbeing of Medicare beneficiaries.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

To encourage patient and caregiver interest in digital health products, CMS and its partners can focus on a few key strategies:

- **Education and Awareness.** Many patients and caregivers just don't know what's available or how it can help them. CMS can work with providers, community organizations, and health plans to spread the word through simple, relatable messaging that could be delivered via workshops at clinics, social media campaigns, and printed guides in plain language.
- **Build Trust Through Providers.** People are more likely to try something new if their doctor or nurse recommends it. CMS can incentivize providers to introduce digital tools during appointments, especially for chronic disease management or post-hospital care.
- **Ease of Use and Accessibility.** Digital tools should be easy to set up and use, especially for older adults or people with limited tech experience. CMS can partner with tech companies to ensure products are user-friendly and offer support like live help or how-to videos.
- **Address Cost and Coverage.** If patients think they have to pay out of pocket, they may not even consider using digital tools. CMS should clearly communicate what's covered and look for ways to expand access to low-cost or no-cost digital solutions under Medicare and Medicaid.
- **Cultural Relevance and Inclusion.** Digital tools should reflect the variety of the populations who use them. CMS and partners can prioritize tools that support multiple languages and are designed with input from communities of color, rural populations, and people with disabilities.
- **Demonstrate Value.** Sharing real stories about how these tools have helped others-especially peers with similar health challenges can go a long way. CMS can support campaigns or peer ambassador programs that spotlight these success stories.

CMS and its partners should work to increase patient and caregiver engagement with digital health products by focusing on accessibility, education, and incentives. These are strategies to encourage adoption:

1. **Improving Awareness and Education.** CMS seeking public input on how to make digital health tools more user-friendly and accessible for Medicare beneficiaries.
2. **Enhancing Interoperability.** Enhance efforts like Blue Button 2.0 and the Interoperability and Patient Access Rule aim to make health data easier to access and share.
3. **Eliminating Barriers to Use.** Explore ways to simplify digital health enrollment and streamline authentication for patients and caregivers.
4. **Expanding Coverage for Digital Therapeutics.** Enhance the Access to Prescription Digital Therapeutics Act to increase Medicare reimbursement for software-based treatments.

5. **Encouraging Innovation.** Work with health tech developers to create secure, patient-centered digital health solutions.

a. **What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?**

CMS's Role in Reviewing Digital Health Products

CMS should consider establishing a voluntary review framework for digital health products that evaluates efficacy, quality, and impact on health outcomes. This framework would not replace regulatory approval or coverage decisions but would act as a trusted resource for stakeholders. The primary role of CMS would be to create transparency, set expectations for high-quality digital tools, and support adoption by signaling which tools are likely to improve care delivery and patient experience.

By providing structured guidance and public-facing evaluation criteria, CMS can help developers align their products with clinical, ethical, and interoperability standards. Additionally, CMS can create a pathway for innovation that is responsive to the needs of patients, caregivers, and providers without creating regulatory barriers.

Criteria for a Voluntary Review Process

- **Clinical Efficacy.** CMS could require digital health tools to present evidence of clinical benefit. This might include peer-reviewed studies, outcomes from randomized controlled trials, or real-world data showing measurable health improvements. For example, a remote monitoring tool for heart failure should demonstrate reduced readmissions or improved medication adherence.
- **Patient-Centered Design.** Tools should be developed with direct input from patients and caregivers. This includes testing the product with focus groups of intended users to ensure ease of use, clear instructions, and accessibility. Features like large fonts, voice-guided prompts, and simple user interfaces can increase engagement, especially for older adults or individuals with disabilities.
- **Data Privacy and Security.** Products must follow strict privacy and security standards. This means encryption of personal health information, clear consent mechanisms, and compliance with HIPAA regulations. Developers should also be transparent about how data is used, stored, and shared, and they should provide patients with control over their information.
- **Interoperability**. Digital health products should be able to exchange data with other systems using established standards such as FHIR (Fast Healthcare Interoperability Resources). This enables providers to incorporate data from the tools directly into clinical workflows, which improves coordination and reduces administrative burden.

- **Equity and Inclusion.** CMS should evaluate whether the product addresses disparities in health outcomes or access. Products that serve multilingual populations, rural residents, or historically underserved groups should be prioritized. Developers should include equity assessments during product development and implementation.
- **Usability and Engagement**. A tool's effectiveness is tied to how consistently patients and caregivers use it. CMS should assess user adoption, frequency of use, and engagement over time. Data on user satisfaction, completion rates of digital tasks, and dropout rates can help gauge usability.
- **Impact on Costs and Utilization.** Tools that help reduce unnecessary healthcare utilization, such as emergency department visits or hospital re-admissions, should be favorably reviewed. CMS may also consider tools that streamline administrative workflows or reduce provider burnout as part of their cost-efficiency evaluation.

Technology Solutions to Boost Adoption

- **Improved Data Access.** Patients need better access to their own health data across systems and applications. CMS can support this by promoting digital infrastructure that enables patients to download and share their data securely. APIs, patient portals, and app-based access should be standardized and widely available.
- **Digital Literacy Support.** Not all patients or caregivers are comfortable with technology. CMS could support community-based digital literacy programs that teach users how to set up, navigate, and troubleshoot digital health tools. These efforts can be especially valuable for seniors and those in low-resource communities.
- **Integration with Clinical Workflow.** Digital tools should not create more work for clinicians. Instead, they should complement existing processes by integrating into electronic health records or care coordination platforms. CMS can encourage the development and adoption of tools that minimize disruption to daily provider routines.

Policy Changes to Encourage Use

- **Value-Based Payment Incentives.** CMS can encourage the use of digital tools by integrating them into value-based payment models. Providers should be rewarded for using digital health products that improve outcomes, enhance patient engagement, or reduce costs. For instance, remote patient monitoring could be reimbursed under chronic care management if it leads to fewer complications.
- **Streamlined Innovation Pathways.** CMS can simplify the process for incorporating digital tools into Medicare and Medicaid. This could include clearer pathways for developers to demonstrate value, templates for contracting, and guidance on how to collect the right types of data to support future coverage consideration.
- **Grant and Demonstration Programs.** Federal funding can help providers and innovators pilot new digital tools, especially in underserved communities. These programs can help generate evidence, test scalability, and identify best practices before wider adoption. CMS can use demonstration results to inform future policy updates.

Program Design Enhancements

- **Beneficiary Education Campaigns.** CMS can fund outreach campaigns that explain the benefits of digital health tools in ways that are easy to understand and culturally relevant. These campaigns can use TV, radio, social media, and community-based events to reach different demographics. Real-life stories and testimonials can help build trust and interest.
- **Provider Enablement.** Clinicians are often gatekeepers of new technology. CMS should invest in training and resources to help providers feel confident in recommending digital tools to patients. Educational toolkits, continuing education modules, and integration guides can support this effort.
- **Feedback Loops and Continuous Improvement.** CMS can create platforms that allow patients, caregivers, and providers to report on their experience with digital tools. This feedback should inform ongoing updates to evaluation criteria and policy guidance. It will also help developers make improvements and maintain alignment with user needs.

Conclusion. CMS has an opportunity to be a trusted partner in the digital health ecosystem by promoting transparency, quality, and innovation without becoming a regulator. Through a voluntary review framework, clear evaluation standards, targeted technology and policy support, and robust outreach efforts, CMS can help accelerate the adoption of digital health products that truly improve care for patients and caregivers.

b. What changes would enable timely access to high quality CMS and provider generated data on patients?

Timely access to high quality CMS and provider-generated data is essential for informed decision making, rapid innovation, and improved patient outcomes. To achieve this goal, the healthcare ecosystem needs coordinated changes in policy, technology, program design, and incentives.

1. Policy and Regulatory Reforms

- **Mandated Timeliness for Claims and Encounter Data**
  CMS could require that claims and encounter data be submitted within a shorter timeframe—such as 30 days instead of the current 90-day window. This would reduce the lag in available data and allow analyses to reflect more recent patient activity.
- **Standardized Data Reporting Requirements**
  Establish clear guidelines on which data elements providers must share, including patient demographics, diagnoses, procedures, lab results, and medication details. By defining a minimum dataset and requiring standardized formats, CMS can reduce variability and improve data completeness.
- **Enhanced Enforcement of Existing Interoperability Rules**
  Strengthen the enforcement of the 21st Century Cures Act rules around information blocking. Providers who fail to share data in a timely manner should face financial or reputational penalties, while those who comply receive recognition or incentives.

2. Technology Solutions and Infrastructure

- **Universal Adoption of FHIR-Based APIs**
  Encourage or require all providers and vendors to implement Fast Healthcare Interoperability Resources (FHIR) APIs for sharing patient data. This would allow authorized systems to query real-time records rather than rely on batch file transfers.
- **Real-Time Data Feeds and Streamlined Data Pipelines**

Invest in technology that supports continuous data streaming from EHRs to CMS databases. For example, providers could send clinical event notifications—such as admissions or discharges—immediately after they occur. This real-time exchange would reduce delays and support more up-to-date analytics.

- **Health Information Exchange (HIE) Expansion and Modernization**
  Expand the reach of regional and statewide HIEs to cover more providers, particularly in underserved or rural areas. Modernize these exchanges to handle high-volume data transfers securely and efficiently. Integration with CMS systems would allow data from multiple sources to flow into centralized analytics platforms.

3. Program Design and Incentive Structures

- **Value-Based Payment Models Tied to Data Sharing**
  Link portions of provider reimbursement to demonstrated data sharing performance. For example, providers who achieve a high percentage of timely, complete data submissions could receive bonus payments or higher quality scores.
- **Grant Programs for Infrastructure Upgrades**
  Offer targeted grants or technical assistance to smaller practices and rural health centers to upgrade their EHRs, invest in API integrations, and train staff on data exchange protocols. By lowering the financial barrier, CMS can accelerate the adoption of necessary technology.
- **Public–Private Data Collaboratives**
  Create consortiums where CMS, major provider organizations, payers, and health IT vendors agree on shared governance, data standards, and privacy safeguards. These collaboratives can pilot innovative data-sharing models that, once proven, can be scaled broadly.

4. Data Governance and Quality Assurance

- **Rigorous Data Validation and Feedback Loop**
  Implement automated checks for data completeness, accuracy, and consistency before data are ingested into CMS analytic systems. Provide rapid feedback to providers when data does not meet quality thresholds so they can correct issues quickly.
- **Standardized Terminologies and Code Sets**
  Require use of uniform code sets—such as SNOMED CT for clinical terms and LOINC for lab results—to ensure that data from different providers can be compared and aggregated without extensive mapping efforts.
- **Transparent Data Quality Metrics**
  Publish dashboards that show data timeliness and quality metrics by provider or region. Transparency can motivate providers to improve their data submission processes and help researchers identify data gaps.

5. Privacy, Security, and Patient Consent

- **Streamlined Consent Management**
  Develop a standardized consent framework that allows patients to opt in or out of data sharing through their patient portals. Clear explanations of how data will be used, as well as the benefits to their care, can increase patient willingness to opt in.
- **Robust Encryption and Access Controls**

Ensure all data pipelines between EHRs, HIEs, and CMS systems use end-to-end encryption. Implement role-based access controls so that only authorized personnel can view identifiable patient data.

- **Regular Security Audits and Compliance Checks**
  Conduct periodic audits of data exchange networks and provider systems to verify adherence to HIPAA and other security standards. Any vulnerabilities should be addressed promptly to maintain trust in the system.

6. Patient Access and Empowerment

- **Unified Patient Portals with Downloadable Records**
  Encourage providers to link local portals to a centralized CMS portal where patients can view and download their combined CMS and provider data. This empowers patients to share their information with care teams, caregivers, or third-party digital health apps.
- **User-Friendly Data Summaries and Interpretations**
  Provide patients with easy-to-understand summaries of their health data, such as trend graphs for lab results or medication refill reminders. When patients see the value of having timely data, they may be more willing to update their information and use digital tools.
- **Educational Campaigns on Data Rights and Benefits**
  Launch outreach efforts that explain how patient-generated and provider-generated data can improve care coordination, reduce redundant tests, and support research. Highlight success stories where timely data access led to better health outcomes.

7. Collaborative Governance and Continuous Improvement

- **Stakeholder Advisory Councils**
  Form committees consisting of patients, caregivers, providers, payers, and health IT experts to guide CMS on evolving data needs and challenges. Regular meetings can surface real-world issues and inform timely policy adjustments.
- **Pilot Testing and Phased Rollouts**
  Before nationwide implementation, pilot new data-sharing requirements in select regions or provider networks. Use lessons learned to refine workflows, identify technical hurdles, and adjust incentives before scaling up.
- **Ongoing Monitoring and Evaluation**
  Establish key performance indicators—such as average data latency, data completeness rates, and provider adoption percentages—and track them over time. Use these metrics to measure the impact of each reform and adjust strategies accordingly.

Conclusion. Achieving timely access to high quality CMS and provider-generated data requires a multifaceted approach. By enacting policy reforms that mandate faster data submissions, investing in technology infrastructure for real-time exchanges, aligning incentives through payment models, strengthening data governance, safeguarding patient privacy, empowering patients, and fostering collaborative governance, CMS can create an ecosystem in which data flows more freely and accurately. These changes will support innovation, improve care coordination, and ultimately lead to better health outcomes for beneficiaries.

***PC-6.* What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?**

For Medicare beneficiaries and caregivers, especially those with limited experience using digital technology, accessibility and simplicity are essential. Digital health products should be designed to meet the specific needs of older adults, individuals with chronic conditions, and users who may face physical, cognitive, or social barriers. The following features are most important for ensuring usability and adoption:

1. Intuitive, Simple Interfaces

- Clean, uncluttered layouts with large buttons and easy-to-read fonts help reduce cognitive load.
- Consistent navigation patterns and visual guidance help users' complete tasks such as checking results or messaging providers without confusion.
- Step-by-step prompts or guided tutorials can make platform adoption less intimidating.

2. Multimodal Access Options

- Users should be able to interact with tools through touch, voice commands, or keyboard input.
- Compatibility with smartphones, tablets, and desktops allows users to choose their preferred device.
- Voice-based navigation or screen readers improve usability for those with visual impairments or dexterity challenges.

3. Language and Literacy Support

- Tools should provide content in multiple languages with easy-to-understand wording.
- Audio explanations, large icons, and visual learning aids support users with limited reading skills.
- A reading level around sixth grade or below helps ensure comprehension across diverse populations.

4. Personalization and Flexibility

- Users should be able to customize font size, contrast, notification settings, and display modes to fit their comfort.
- Personalized health goals, reminders, and condition-specific guidance increase relevance and engagement.
- Providing a simplified user mode for new users and an advanced option for experienced ones allows flexibility without overwhelming beginners.

5. Human Support and Onboarding Assistance

- Access to real-time support from care navigators or helplines helps new users get started with confidence.
- In-person onboarding or remote walkthroughs can build trust and minimize frustration.
- Caregiver accounts that allow family members to assist with navigation are especially important for users with limited digital experience.

6. Integration with Existing Care Relationships

- Digital tools should align with care already being delivered by providers and not function in isolation.
- Features that enable direct communication with providers or access to visit summaries support continuity and trust.
- Tools that are introduced and recommended by providers are more likely to be accepted by patients.

7. Clear Privacy and Security Information

- Tools should explain how data is protected using plain language and visual aids.
- Consent forms should be easy to understand and offer clear explanations of what data is shared and why.
- Building trust in the system is essential for long-term engagement.

8. Offline Capabilities and Low Bandwidth Support

- Not all users have access to fast or consistent internet connections.
- Products should offer offline functionality or operate smoothly on low-bandwidth networks.
- Downloadable guides, asynchronous messaging, and offline tracking features help bridge connectivity gaps.

9. AI-Powered User Guidance and Support

- Incorporating AI assistants into digital health tools can dramatically enhance usability, especially for first-time users.
- AI can offer real-time guidance, suggest the next steps, and provide natural language answers to user questions.
- For example, a voice-enabled virtual assistant could walk a user through checking lab results, scheduling an appointment, or learning about a new medication.
- These AI features should be friendly, non-technical, and responsive to user behavior, offering support without judgment.
- When combined with human support, AI-guided assistance creates a hybrid experience that improves confidence and reduces the learning curve.

10. Regular Feedback and User Testing

- Tools should be tested by real Medicare beneficiaries and caregivers early and often.
- Built-in feedback options help developers continuously improve their experience based on real user needs.
- CMS could also provide benchmarks or evaluation standards for usability, informed by patient and caregiver feedback.

Conclusion. To be effective for Medicare beneficiaries and caregivers, digital health products must be built for simplicity, guidance, and inclusion. By combining intuitive design, personalized support, flexible access, and AI-powered assistance, developers can create tools that empower users rather than overwhelm them. These features not only enhance adoption but also improve health outcomes by helping patients stay engaged in their care. As digital tools continue to evolve, the inclusion of human-centered design and intelligent guidance will be key to ensuring no one is left behind.

***PC-7.* If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?**

If CMS were to collect real-world data on the impact of digital health products after-market release, it would be important to use a multi-source, standardized, and scalable approach. The goal should be to capture data that reflects actual use in diverse populations and care settings, while maintaining rigor and minimizing burden on providers and patients.

1. Leverage Existing CMS Data Infrastructure

- **Claims and Encounter Data**

CMS already collects claims and encounter data across Medicare, Medicaid, and other programs. These sources can be used to track utilization patterns, health outcomes, and cost trends related to digital health product use. For example, reductions in hospital readmissions, emergency visits, or length of stay can be analyzed over time.

- **Integration with Chronic Care and Innovation Models**
  Digital health product use can be embedded as a variable in ongoing models such as ACO REACH, BPCI Advanced, or CMMI pilots. These models already collect longitudinal outcomes and cost data.

2. Establish a Voluntary Digital Product Registry

- Create a centralized registry where manufacturers, providers, and researchers can submit data on digital product use, patient characteristics, and observed outcomes.
- Registries should use standard data formats and align with FHIR and USCDI to support interoperability.
- CMS could provide incentives, technical support, or regulatory advantages for participation.

3. Enable EHR-Integrated Data Collection

- Encourage or require integration of digital product usage data into certified EHRs using APIs.
- Through partnerships with providers and health systems, CMS can gain access to real-time clinical data such as vitals, lab results, medication adherence, and care plan milestones that are influenced by digital tool usage.
- Structured data fields can track specific interventions and outcomes tied to the digital tool being studied.

4. Incorporate Patient-Reported Outcomes and Experience Measures

- Use patient-reported outcome measures (PROMs) and patient-reported experience measures (PREMs) to assess quality of life, functional status, satisfaction, and usability.
- This data can be collected through the digital tools themselves or via follow-up surveys administered through patient portals or care teams.
- Collecting feedback from caregivers is also essential, especially when they are primary users of technology.

5. Utilize AI and Advanced Analytics to Detect Signals

- Apply AI and machine learning to large datasets to identify patterns in outcomes and costs associated with digital tool adoption.
- This can help isolate effects of specific tools or features, adjusting for confounders like population risk or socioeconomic factors.

6. Collaborate with Health Information Exchanges and State Medicaid Programs

- HIEs can provide access to a broader range of real-world clinical data, including services not billed to CMS directly.
- State Medicaid programs can serve as early testing grounds for data-sharing approaches, especially in vulnerable populations.

7. Build a Framework for Ongoing Post-Market Surveillance

- Develop a systematic process for ongoing monitoring, similar to the FDA's Sentinel Initiative but focused on digital health.
- Include safety, effectiveness, equity, and cost-related endpoints.
- Use tiered levels of evidence to inform potential changes in coverage, quality metrics, or reimbursement over time.

8. Encourage Manufacturer Partnerships for Evidence Sharing

- Require or encourage digital health developers to share anonymized data on user outcomes from commercial implementations.
- This can include dashboard metrics, engagement data, and adverse event reports that are difficult to obtain through claims alone.

9. Include Equity-Focused Metrics

- Collect and stratify data by race, ethnicity, language, income, geography, and disability status.
- This helps CMS understand whether digital products are narrowing or widening disparities in access, outcomes, and costs.

Conclusion. To effectively collect real-world data on digital health products, CMS should combine its existing data infrastructure with new methods that capture real-time, patient-centered, and context-aware information. By integrating clinical, claims, and patient-reported data, encouraging cross-sector collaboration, and applying modern analytics, CMS can evaluate the true impact of digital health tools on care quality, outcomes, and spending. This approach supports responsible innovation and helps ensure that digital transformation delivers measurable value to patients and the health system.

2. Data Access and Integration

***PC-8*. In your experience, what health data is readily available and valuable to patients or their caregivers or both?**

a. **What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?**

Any health app, portal, or caregiver workbook should be able to pull down a patient's complete record, structured and unstructured, upon patient approval. That means not just a PDF of yesterday's lab results, but live feeds of everything from past claims to tomorrow's appointment slots.

1. Readily Available and High-Value Data

These elements are now routinely exposed via EHR portals, FHIR® APIs, or device-vendor SDKs—so most apps and caregivers can pull them in real time:

- Structured EHR Data
  - Labs, vitals, problem lists, immunizations, procedure codes
  - Discrete medication lists (name, dose, frequency) and allergy records
- Discharge Summaries and Care Plans
  - Often pushed to patient portals within 24 hours of an inpatient stay
- Basic Scheduling/Referral Info
  - Next appointment time, location, referring provider
- Some Patient-Generated Health Data (PGHD)
  - When patients use apps or wearables that explicitly integrate via FHIR® Device or Observation resources

2. Valuable, but Difficult, Data to Unlock

These data types can transform care planning and cost transparency, but they're still hard for patients, caregivers, or third-party developers to get in a reliable, standards-based way:

- Claims and Cost Data
  - Detailed claim-line items, adjudication notes, prior-authorization decisions
  - Real-time patient out-of-pocket estimates or negotiated price schedules
- Unstructured Clinical Notes
  - Encounter narratives, operative reports, pathology findings
  - Typically buried in free-text and not routinely pushed via FHIR® Composition or Document Reference
- Full Appointment and Resource Calendars
  - Multi-site scheduling across health systems (OR blocks, imaging slots, telehealth links)
- Granular Insurance Coverage Rules
  - Benefit limitations, network restrictions, step-therapy rules
- Rich Wearable and Remote Monitoring Streams
  - Continuous glucose monitor curves, high-resolution ECG strips, sleep-stage epochs
  - Often vendor-locked or only exportable via proprietary portals

3. Why These Gaps Matter

- Care coordination breaks down if every system exposes only part of the picture.
- Shared decision-making suffers when cost estimates and coverage rules aren't at patients' fingertips.
- Innovation stalls when app developers must build custom scrapers or manual uploads instead of using a single API.

4. Next Steps Toward Full PC-8 Compliance

- Push all payers to open up claims and coverage APIs (e.g., via CARIN, USCDI v4)
- Require FHIR® Document Reference for every clinical note and operative report
- Standardize price-transparency endpoints under CMS Rule "Know before You Owe"
- Make multi-system scheduling available via a single Care Coordination FHIR® resource

With those walls down, patients, caregivers, and innovators can finally assemble the 360° view of health and turn into better decisions, smoother transitions, and lower costs for everyone.

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?

These specific sources would be of highest value by filling gaps in care coordination, personalized support, and preventive health.

1. Administrative and Eligibility Data
   Sources: Enrollment files, program-benefit rosters, demographic databases (e.g., Medicaid/Medicare eligibility, local health–social service enrollment)
   Why Valuable:
   - Pinpoints coverage gaps or lapsed benefits so caregivers can avert unexpected bills.
   - Flags high-risk subpopulations (e.g., dual-eligible, low-income) trigger social-needs screening or care-manager outreach.

2. Disease and Procedure Registries
   Sources: State cancer registries (SEER), immunization registries, joint-replacement registries, rare-disease consortia
   Why Valuable:
   - Longitudinal follow-up on outcomes (e.g., five-year survival, prosthesis failure rates), helping patients weigh treatment options.
   - Benchmarks individual progress against population norms (e.g., post-op infection rates).
3. Standardized Patient Assessments
   Sources: Functional status surveys (Katz ADL, PROMIS), mental-health screens (PHQ-9, GAD-7)
   Why Valuable:
   - Quantifies quality-of-life and daily living limitations, enabling caregivers to tailor home support.
   - Tracks symptom trajectories remotely, triggering alerts before crises occur.
4. Patient-Reported Outcomes and Experience Measures
   Sources: CAHPS surveys, disease-specific PRO instruments (e.g., Asthma Control Test), custom digital symptom journals
   Why Valuable:
   - Captures subjective well-being and treatment burdens that never show up in clinical notes—crucial for shared decision-making.
   - Empowers patients to report side-effects or adherence barriers in real time.
5. Chief-Complaint and Triage Logs
   Sources: Hospital/ambulatory registration systems, telehealth chat transcripts
   Why Valuable:
   - Early warning for outbreaks (e.g., flu-like symptoms in a ZIP code).
   - Help caregivers anticipate ED or urgent care needs before a formal diagnosis.
6. Social-Determinants and Community Data
   Sources: Census-based indices (Area Deprivation Index), SNAP/WIC enrollment, housing-assistance records, transportation voucher programs
   Why Valuable:
   - Illuminates' food, housing, or transportation insecurity that drive missed appointments and poor outcomes.
   - Guides referrals to community partners (food banks, ride services) for wraparound support.
7. Environmental and Geospatial Exposures
   Sources: EPA air-quality monitors, local pollen counts, neighborhood "walkability" scores
   Why Valuable:
   - Correlates pollution peaks or allergen spikes with asthma attacks or COPD exacerbations.
   - Empowers caregivers to adjust activity plans or air-filtering strategies proactively.
8. Genomic and Biomarker Repositories
   Sources: Consumer genetic testing (23andMe), clinical labs' biobanks, public resources (NIH All of Us)
   Why Valuable:

- Reveals pharmacogenomic risks (e.g., slow-metabolizer status) to avoid adverse drug reactions.
- Enables precision-prevention strategies (e.g., enhanced screening for high-risk variants).

9. Pharmacy Dispensation and Adherence Data
Sources: National PDMPs, pharmacy point-of-sale systems, smart-pill bottle logs
Why Valuable:
- Confirms whether a prescription was actually filled—and refilled on schedule—so caregivers can intervene on nonadherence.
- Detects early signs of potential misuse or diversion.

By weaving these diverse sources into a single, patient-authorized portal or app, caregivers gain a 360° view of needs—medical, social, and environmental—so they can anticipate risks, tailor interventions, and keep patients safer and more engaged.

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

Regulatory, technical leadership, and market ownership are readily available to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence. The standardization of administrative and technical content relating to clinical research and generation of actionable evidence among stakeholders is required.

Standardizing Data. There are efforts to standardize clinical research data and evidence generation to improve accuracy, reliability, and interoperability. For example, CDC's Clinical Standardization Programs (CSP)[i] focus on ensuring laboratory measurements are consistent across different systems, laboratories, and timeframes. This helps healthcare providers make evidence-based decisions with confidence.

Additionally, global initiatives aim to standardize clinical research data to enhance traceability, accessibility, and reproducibility. The International Council for Harmonization of Technical Requirements for Pharmaceuticals for Human Use (ICH) has worked on harmonizing regulatory submissions for new product approvals.[ii]

In medical affairs, integrated evidence-generation strategies are becoming more important. These strategies combine traditional clinical trials with real-world evidence (RWE) to provide more actionable insights for healthcare providers, regulators, and payers.[iii]

Electronic Health Records (EHR). The National Trends in Hospital and Physician Adoption of Electronic Health Records (EHR) is good. As of 2021, nearly 4 in 5 office-based physicians (78%) and nearly all non-federal acute care hospitals (96%) adopted a certified EHR. This marks substantial 10-year progress since 2011 when 28% of hospitals and 34% of physicians had adopted an EHR.[iv]

Accessibility, interoperability, and integration have increased due to the laws addressing these topics. Additionally, The Department of Health and Human Services issued regulations and policies supporting accessibility, interoperability, and integration.

Certified EHR Technology. In order to efficiently capture and share patient data, health care providers need certified electronic health record (EHR) technology (CEHRT) that stores data in a structured format. Structured data allows health care providers to easily retrieve and transfer patient information and use the EHR in ways that can aid patient care.[v]

CMS and the Office of the National Coordinator for Health Information Technology (ONC) have established standards and other criteria for structured data that EHRs must meet to qualify for use in the Medicare Promoting Interoperability Program.

Most clinical information is digitized, accessible, and shareable thanks to several technology and policy advances making interoperable, electronic health record systems widely available. In 2016, the 21st Century Cures Act (Cures Act) made sharing electronic health information the expected norm in health care and authorized the Secretary of Health and Human Services (HHS) to identify "reasonable and necessary activities that do not constitute information blocking." Information blocking exceptions are identified in 45 CFR Part 171. Learn more about laws, regulations, and policies related to information blocking.[vi]

What Is Information Blocking and to Whom Does It Apply?

Information blocking is a practice by an "actor" that is likely to interfere with the access, exchange, or use of electronic health information (EHI), except as required by law or specified in an information blocking exception. The Cures Act applied the law to healthcare providers, health IT developers of certified health IT, and health information exchanges (HIEs)/health information networks (HINs).

It is also important to note that the Cures Act established two different "knowledge" standards for actors' practices within the statute's definition of "information blocking." In particular, for health IT developers of certified health IT, as well as HIEs/HINs, the law applies the standard of whether they know, or should know, that a practice is likely to interfere with the access, exchange, or use of EHI. For healthcare providers, the law applies the standard of whether they know that the practice is unreasonable and is likely to interfere with the access, exchange, or use of EHI.

Health care providers participating in the Medicare Promoting Interoperability Program must attest to certain statements to demonstrate that they have not knowingly and willfully taken action to limit or restrict the compatibility or interoperability of their CEHRT.

The Opportunities

- **Community-Adopted Data and Metadata Standards.** Establishing and enforcing common data elements (CDEs), standardized terminologies (LOINC, SNOMED), and rich metadata schemas makes it vastly easier to pull data from multiple EHRs and devices for both primary and secondary use.
- **Real-Time, API-Driven Exchange.** Moving from hourly or “overnight” batch feeds to event-driven FHIR® interfaces or streaming architectures ensures researchers and clinicians work with the freshest data—critical for safety monitoring and near-real-time evidence generation.
- **Unlocking Unstructured Clinical Narratives.** Requiring every system to expose notes, operative reports, and pathology findings via FHIR® Document Reference or Composition

resources would turn buried free text into mineable signals for outcomes research and NLP-powered phenotyping.

- **Cloud-Native Data Lakes and Analytics Platforms.** Leveraging scalable, secure cloud infrastructures and enterprise data-warehouse patterns unify disparate datasets under one governed environment, lowering the complexity and cost of spinning up research cohorts or running advanced AI/ML jobs.
- **Federated Research Networks and Common Data Models.** Networks like PCORnet, All of Us, and OHDSI's OMOP model let institutions keep data behind local firewalls while enabling aggregate queries or distributed analyses—balancing privacy and collaboration without a single monolithic repository.

The Challenges

- **Fragmented Standards Adoption and Vendor Lock-In.** EHRs and specialty systems frequently implement "their own flavor" of FHIR or proprietary formats, forcing custom adapters and eroding true plug-and-play interoperability.
- **Variable Data Quality and Completeness.** Missing fields (e.g., problem lists), inconsistent coding practices, and lack of provenance metadata undermine the validity of cross-site research.
- **NLP and Annotation Overhead for Free Text.** Turning encounter notes and operative reports into structured, research-ready data still demands expensive NLP pipelines and human curation.
- **Governance, Consent and Privacy Hurdles.** Fragmented IRB processes, complex Data Use Agreements, and evolving privacy laws make multi-institutional data pooling a legal and administrative maze.
- **Limited Real-Time Feeds and Context Loss.** Many systems still rely on periodic syncs; even when data arrives, clinical context (why a lab was ordered, encounter relationships) often vanishes in translation.
- **Organizational and Cultural Silos.** Hospitals, payers, and academic centers may see data as a competitive asset, resisting open sharing without strong incentives or mandates.
- **High Cost and Technical Complexity.** Stand-up of API infrastructures, data-cleaning pipelines, and secure enclaves requires significant investment in talent and tooling—often beyond the budget of smaller organizations.

***PC-9.* Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?**

Below is a list of CMS data sources that would turn a basic Blue Button 2.0 feed (demographics + Part A/B/D claims) into a powerhouse for patient- and caregiver-facing apps. Richer data feeds to power next-gen digital tools including:

1. Medicare Advantage (Part C) Encounter and Payment Data

   a. Why it matters: Nearly half of Medicare enrollees are in MA plans. Without encounter-level data (services, costs, utilization), apps can't give MA members a full picture of their care or out-of-pocket spend.
2. Provider and Facility Directory + Quality Metrics
   a. PECOS provider rosters, Hospital Compare/Physician Compare star ratings, telehealth offerings, specialties and office hours.
   b. Why it matters: Empowers patients to find, evaluate and book with high-quality clinicians in their network.
3. Plan Benefit Details, Formularies and Star Ratings (Parts C and D)
   a. Drug tiering, deductibles, coverage gaps, pharmacy networks, plan-year star ratings.
   b. Why it matters: Enables apples-to-apples plan comparisons and real-time cost-estimate engines so patients can choose, or switch plans based on personalized drug and service needs.
4. Prior-Authorization and Utilization Management Logs
   a. Real-time request/approval/denial data for imaging, procedures, therapies.
   b. Why it matters: Shines light on which services need pre-approval, how long it takes, and why requests are denied—so caregivers can plan around barriers.
5. Chronic Condition Flags and Risk Adjustment (CCW)
   a. Chronic Condition Warehouse indicators, Hierarchical Condition Category (HCC) risk scores.
   b. Why it matters: Lets apps stratify patients by disease burden and predicted cost, triggering tailored coaching, alerts or care-manager outreach.
6. Home Health, Hospice and Long-Term Care Outcomes
   a. OASIS assessments, hospice quality metrics, SNF and nursing-home star ratings.
   b. Why it matters: Gives caregivers insight into post-acute recovery, staffing levels and facility performance—critical for safe discharge planning.
7. Appeals, Grievances and Coverage Determinations
   a. CMS appeals/denial records, Explanation of Medicare Benefits (EOMB) appeals data.
   b. Why it matters: Equips patients and advocates to track, manage and contest coverage decisions in one place rather than juggling paper notices.

a. What difficulties are there in accessing or utilizing these data sources today?

Below is a breakdown of why each "missing" CMS data feed is so hard for patient- and caregiver-apps to get today—and the cross-cutting barriers that keep them siloed.

1. **Medicare Advantage (Part C) Encounter and Payment Data**
   - Fragmented reporting – MA plans submit encounter records at varied intervals (quarterly, semi-annual) and often via non-FHIR formats.
   - Quality and completeness issues – CMS historically suppresses or aggregates small-cell counts to protect plan confidentiality, so apps see holes or delays.
   - DUA gating – Access requires a strict Data Use Agreement for MA encounter data, plus technical on-boarding that many small vendors can't afford.
2. **Provider and Facility Directory + Quality Metrics**

- Stale or inaccurate rosters – PECOS and Provider Compare feeds can be out-of-date by months. Phone hours, telehealth URLs, even specialty flags are often missing.

- No unified, standards-based API – Facility Compare and Physician Compare exports come as CSV/Excel dumps, not FHIR Location/Practitioner resources, so you must build custom parsers.
- Inconsistent identifiers – NPI vs. CCN vs. Taxonomy codes don't always map neatly, breaking automated lookups.

3. **Plan Benefit Details, Formularies and Star Ratings (Parts C and D)**

- PDF-locked details – Formularies, benefit booklets, drug tier tables still ship as PDFs, not machine-readable Formulary FHIR resources.
- Coverage-gap logic complexity – Part D LIS phases and deductible phases require intricate, plan-specific rules that apps must reverse-engineer.
- Star-rating updates only yearly – Dynamic changes (e.g., mid-year formulary tweaks) aren't surfaced until the next plan year.

4. **Prior-Authorization and Utilization Management Logs**

- No public API or data feed – PA decisions live in payer portals or faxed letters, not in FHIR ClaimResponse or CoverageRequirementResult.
- Manual, paper-based workflows – Denials/approvals get communicated by phone or letter, meaning apps must rely on patient self-reports.
- Lack of granularity – Even if you scrape it, reasons for denial are free-text, unstructured, and hard to standardize.

5. **Chronic Condition Flags and Risk Adjustment (CCW, HCC)**

- Behind-the-firewall data warehouse – CCW indicators and HCC scores sit in CMS's Research Data Assistance Center (ResDAC), requiring lengthy DUA approvals.
- Black-box scoring – HCC risk models change year to year, so even if you get the data, reproducing up-to-date risk scores is nontrivial.
- Batch-only extracts – You can't query "my patient's current HCC score" in real time—only download large annual tables.

6. **Home Health, Hospice and Long-Term Care Outcomes**

- OASIS assessments under DUA – The detailed home-health OASIS item-level data isn't in Blue Button; it lives in a restricted CMS data set.
- Quality metrics lag – Nursing Home Compare releases star ratings quarterly, and API endpoints are still emerging.
- Disparate sources – You must stitch together OASIS, Hospice Compare, SNF Quality Reporting Program feeds, each with different schemas.

7. **Appeals, Grievances and Coverage Determinations**

- Scattered records – Appeals live in the Medicare Beneficiary Database; grievances in separate Complaint Tracking modules—no unified feed.
- Unstructured text – Explanations of Benefits (EOBs) or appeal letters are often PDF/scanned images, not FHIR Document Reference resources.
- Limited patient-access channels – Beneficiaries receive mailed notices, but there's no guaranteed digital channel to pull that data into an app.

b. What suggestions do you have to improve the Blue Button 2.0 API experience?

Moving Toward a Unified Future

- Push CMS to expand Blue Button into a "BB 3.0" that supports Part C/D plan details, provider directories, and pre-auth logs as first-class FHIR resources.

- Simplify DUAs with a one-stop, patient-signed consent framework that covers all data types.
  Mandate standard FHIR profiles for formulary, prior-auth, OASIS, appeals, etc., under upcoming CMS interoperability rules.
- Incentivize payers and MA plans with financial carrots (bonus payments) to publish real-time, standards-based APIs.

Cross-Cutting Barriers

- **Data Fragmentation** – You need to integrate half a dozen CMS APIs (Blue Button, BCDA, DPC, Compare sites, ResDAC) just to piece together a full picture.
- **Governance and DUAs** – CMS data agreements can take months, require security attestation, and sometimes demand on-premise enclaves.
- **Standards Gaps** – Many critical feeds exist only as bulk CSVs or PDFs, not as FHIR resources with consistent profiles.
- **Technical Overhead** – Small developers lack the engineering bandwidth to build custom ETL pipelines for each feed.
- **Lack of Real-Time Access** – Most CMS data updates in nightly or quarterly batches, making near-patient, just-in-time apps impossible.

With those levers pulled, developers could build true one-stop apps—real-time care coaches, cost-estimate engines, discharge planners—that finally deliver on the promise of patient-centered digital health.

c. Is there non-CMS data that should be included in the API?

Yes, and an API that goes beyond CMS data can offer a much richer picture of a patient's overall health status and journey. Here are some key categories of non-CMS data that could be valuable and why they matter:

1. **Patient-Generated Health Data (PGHD)**
   - **Examples:** Data from wearables, mobile health apps, smart devices (activity tracking, heart rate, blood glucose monitoring, sleep patterns, and mood logs).
   - **Why It Matters:** PGHD provides real-time, continuous insights into a patient's daily health and behavior patterns. It enables personalized health recommendations, proactive monitoring, and early identification of issues—especially useful for chronic disease management.
2. **Electronic Health Record (EHR) Data from Healthcare Providers**
   - **Examples:** Detailed clinical notes, imaging reports, lab results, and medication reconciliation data that are not part of the CMS claims feed.
   - **Why It Matters:** EHR data offers clinical context and richer insights into diagnosis, progression, and treatment outcomes. Access to this information can enable more nuanced care coordination and support clinical decision-making by seamlessly integrating hospital- or clinic-generated records with CMS data.
3. **Provider and Facility Quality and Performance Information**

- **Examples:** Publicly reported star ratings, performance metrics, patient satisfaction scores, and independent provider reviews from organizations like Hospital Compare or third-party accreditation agencies.
- **Why It Matters:** This data allows patients and caregivers to evaluate healthcare providers beyond geographic proximity or network affiliation. It informs provider selection by highlighting quality of care and patient experiences.

4. **Pharmacy and Drug Pricing Data**
   - **Examples:** Formularies, retail pharmacy pricing, drug tier information from pharmacy benefit managers (PBMs), and out-of-pocket cost estimates provided by private market data aggregators.
   - **Why It Matters:** Price transparency and cost estimates empower patients to make informed decisions regarding their prescriptions. This data can also help compare alternatives in situations where cost and access are key considerations.
5. **Social Determinants of Health (SDOH) and Environmental Data**
   - **Examples:** Local socioeconomic indicators, neighborhood-level environmental data (like air or water quality, walkability scores), food access indices, and transportation availability.
   - **Why It Matters:** SDOH and environmental factors play a critical role in overall health outcomes. Integrating this data can help tailor interventions, support risk stratification, and guide community-based care strategies that address broader factors affecting health.
6. **Non-CMS Payer and Insurance Data**
   - **Examples:** Claims and benefit information from private insurers that round out the picture of a patient's overall insurance landscape, including coverage gaps or coordination between multiple plans.
   - **Why It Matters:** With many patients having combinations of public and private coverage, incorporating non-CMS payer data can give a more complete view of care utilization and financial responsibilities, which is vital for whole-person care and cost transparency.

Incorporating these non-CMS data sources alongside CMS data in a unified API would enable a more holistic approach to patient care. It would allow developers to build applications that not only inform patients about historical claims and coverage but also offer actionable insights based on clinical nuances, behavioral patterns, quality/price comparisons, and the broader social context that influences health.

3. Information Blocking and Digital Identity

***PC-14.* Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSP)):**

a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?

The challenges today in getting patients/caregivers to sign up and use digital identity credentials include:

1. **High Costs** – Hospitals and healthcare providers must invest in **secure identity proofing infrastructure**, which can be expensive.
2. **Complex Integration** – Many healthcare systems rely on **legacy EHR platforms** that may not support advanced authentication methods.
3. **Regulatory Compliance** – Meeting **HIPAA, GDPR, and other privacy laws** while implementing **multi-factor authentication (MFA)** can be challenging.
4. **User Adoption and Training** – Healthcare professionals and patients may struggle with **new authentication processes**, leading to resistance.
5. **Remote Identity Proofing Limitations** – Verifying patient identities **without in-person validation** can be difficult, especially for telemedicine.
6. **Cybersecurity Risks** – While MFA enhances security, **phishing-resistant authentication** methods are still evolving.

Hospitals and healthcare providers must balance **security, usability, and compliance** when adopting these standards.

b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used?

Digital identity credentials could bring a wave of improvements to healthcare, making life easier for both patients and caregivers. The healthcare industry is already moving toward digital transformation using identity as over 95.3% of hospitals have adopted Electronic Health Records (EHRs), meaning a significant portion of the population has some form of digital medical identity. Market share can be divided as follows:

1. Epic: 37.7%
2. Oracle Health (Cerner): 21.7%
3. Meditech: 13.2%
4. Evidence (CPSI): 9%
5. Altera Digital Health: 3.4%
6. Medhost: 3.3%
7. WellSky: 2.6%
8. Vista: 2.3%
9. Netsmart Technologies: 2.1%
10. Other: 1.3%

Thus 82% are covered by the top four vendors. 95.3% are covered by the top nine vendors. The wider adoption of digital identity credentials could be a game-changer. Here's how:

- **Better Access to Care** – Patients could securely share their medical history with different providers, reducing paperwork and ensuring continuity of care.
- **Enhanced Privacy and Security** – With decentralized identity solutions, patients would have more control over their health data, reducing risks of breaches and unauthorized access.
- **Reduced Administrative Burden** – Caregivers and healthcare workers could streamline credential verification, cutting down on delays caused by scattered data and inefficient workflows.
- **Improved Efficiency** – Faster authentication processes would allow clinicians to spend more time with patients rather than navigating complex security protocols.
- **Stronger Cybersecurity** – Hospitals and healthcare systems could better protect sensitive patient information while maintaining ease of access for authorized personnel.

c. What are the potential downsides?

While digital identity credentials offer many benefits, they also come with potential drawbacks:

- **Privacy Concerns** – Centralized digital identities could make patient data more vulnerable to breaches or unauthorized access.
- **Access Barriers** – Some patients, especially older adults or those with limited digital literacy, may struggle to navigate digital identity systems.
- **Data Interoperability Issues** – Healthcare providers may use different systems, making it difficult to ensure seamless data sharing.
- **Risk of Identity Theft** – If credentials are compromised, patients could face fraud or misuse of their medical information.
- **Dependence on Technology** – System outages or cyberattacks could disrupt access to critical healthcare services.

Balancing security, accessibility, and efficiency is key to making digital identity credentials work effectively in healthcare.

d. How would encouraging the use of CSPs improve access to health information?

The use of EHR digital identities supplemented by CSP digital identities leverages exist EHR regulatory compliant digital identities. A CMS merged solution is possible where identities are synchronized with EHR infrastructures and CSP infrastructures. The goal is to enhance security, interoperability, and efficiency. Here's how:

- **Secure Authentication** – provide strong identity verification, ensuring that only authorized individuals—patients, caregivers, and healthcare providers—can access sensitive health records.
- **Interoperability Across Systems** – facilitate seamless data exchange between different healthcare platforms, making it easier for patients to access their records across multiple providers.
- **Reduced Administrative Burden** – By streamlining identity verification, cut down on paperwork and manual authentication processes, allowing healthcare professionals to focus more on patient care.
- **Improved Patient Control** – Patients can manage their digital identities securely, granting access to their health data only when necessary.
- **Enhanced Privacy and Compliance** – help healthcare organizations comply with regulations like HIPAA by ensuring that patient data is protected from unauthorized access.

Using CSP infrastructures as supplemental is based on the following usability and accessibility concerns.

- **CLEAR**: Designed for **fast airport security screening** but requires **paid membership**.
- **Login.gov**: Free to use for **government services** but **limited to federal applications**.
- **ID.me**: Used for **government benefits, healthcare, and financial services**, but has faced **privacy concerns** over biometric data collection.

e. What role should CMS/payers, providers, and app developers have in adoption of driving?

Digital identity includes both application identity and user identity. CMS, payers, providers, and app developers all play crucial roles in driving the adoption of digital identity credentials in healthcare:

- **CMS (Centers for Medicare & Medicaid Services).** As a regulatory body, CMS can set standards for digital identity use, incentivize adoption through reimbursement policies, and ensure interoperability across healthcare systems.
- **Payers (Insurance Companies and Health Plans).** Payers can encourage adoption by requiring secure digital identity verification for claims processing, reducing fraud, and streamlining patient authentication.
- **Providers (Hospitals, Clinics, and Physicians).** Healthcare providers can integrate digital identity solutions into their workflows to improve patient access, reduce administrative burdens, and enhance security.
- **App Developers and Health Tech Companies.** Developers can create user-friendly digital identity solutions that prioritize security, accessibility, and interoperability, ensuring seamless integration with existing healthcare infrastructure.

The combined use of digital identity credentials and application identity credentials enhances security by ensuring strong authentication, access control, and fraud prevention.

How these work together for security:

1. Multi-Layered Authentication – Digital identity credentials verify who the user is, while application identity credentials confirm what they have access to. This prevents unauthorized access.
2. Reduced Identity Fraud – Cryptographic verification ensures that credentials are tamper-proof, making it harder for attackers to impersonate users.
3. Secure Access to Sensitive Data – Organizations use verifiable digital credentials to authenticate users before granting access to protected applications and databases.
4. Interoperability Across Systems – Digital credentials allow users to seamlessly authenticate across multiple applications without needing separate logins.
5. Blockchain-Based Verification – Some systems use blockchain technology to ensure credentials remain secure and verifiable without storing sensitive data.

Microsoft Corporation offers authentication to Microsoft Entra ID using Application Identity. Entra Connect uses the Microsoft Entra Connector account to authenticate and sync identities from Active Directory to Entra ID. This account uses username and password to authenticate requests. To enhance the security of the service, they are rolling out an application identity that uses Oauth 2.0 client credential flow with certificate credentials. In this new method, Entra or Administrator creates a single tenant third party application in Entra ID and uses one of the relevant certificate management options below for the credentials. Microsoft Entra Connect provides three options for application/certificate management:

1. Managed by Microsoft Entra Connect (Recommended)
2. Bring Your Own Application (BYOA)
3. Bring Your Own Certificate (BYOC)

By working together, these stakeholders can create a more efficient, secure, and patient-centric healthcare system.

f. How can CMS encourage patients to get digital identity credentials?

CMS can encourage patients to adopt digital identity credentials through several initiatives:

- **Education and Awareness Campaigns.** Informing patients about the benefits of digital identity credentials, such as improved access to health records and enhanced security.
- **Incentives and Rewards.** Offering discounts on healthcare services or faster processing for those who use digital identity credentials.
- **Simplified Enrollment.** Making the process of obtaining digital credentials easy and accessible, especially for older adults and those with limited digital literacy.
- **Integration with Existing Services.** Embedding digital identity verification into Medicare.gov and other CMS platforms to streamline access to healthcare resources.
- **Collaboration with Providers and Payers.** Encouraging healthcare providers and insurers to adopt digital identity solutions, making them a standard part of patient interactions.

RFI Section E. Technology Vendors, Data Providers, and Networks

1. Ecosystem

***TD-1.* What short term (in the next 2 years), and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?**

Developers are key to driving innovation in digital health for Medicare beneficiaries and caregivers. To stimulate their interest, CMS needs to create an environment that makes development attractive, viable, and rewarding. Here are a few key factors that influence developer engagement:

1. **Financial Incentives and Market Opportunity**
   - Developers are more likely to invest time in building digital health tools if they see clear financial incentives. Grant programs, reimbursement models, and partnerships with insurers can make Medicare-focused innovation economically viable.
   - The aging population presents a growing demand for digital health solutions, creating a market opportunity that developers might not have fully explored yet.
2. **Access to High-Quality Data and Developer-Friendly Infrastructure**
   - The easier it is to integrate with Medicare claims data, health records, and existing infrastructure, the more appealing development becomes.
   - CMS can support developers by creating user-friendly APIs, data-sharing policies, and sandbox environments where they can test solutions before going live.
3. **Streamlined Regulatory and Compliance Processes**
   - Medicare reimbursement and regulatory compliance can be complex. Providing clear guidance, reducing bureaucratic barriers, and offering fast-track approvals for digital health tools can make Medicare-focused development more appealing.

4. **Community Building and Collaboration**
   - Hosting hackathons, innovation challenges, and developer workshops can spark creativity and bring new minds into the space.
   - Industry partnerships, especially with tech firms, can expose developers to the Medicare ecosystem and accelerate progress.
5. **Long-Term Support and Sustainability**
   - Developers want reassurance that CMS-backed initiatives will be long-term and scalable.
   - Creating a Medicare digital health marketplace or certification program would provide longevity and trust in solutions built for beneficiaries.

CMS can make Medicare-focused development not only feasible but highly rewarding by focusing on these elements.

TD-2. Regarding CMS Data, to stimulate developer interest-

a. What additional data would be most valuable if made available through CMS APIs?

Integrating electronic health records (EHRs) into CMS APIs would be a game-changer for developers creating digital health solutions for Medicare beneficiaries and caregivers. By enabling seamless interaction between CMS data and EHR systems, developers could build tools that enhance care coordination, improve patient engagement, and optimize clinical workflows.

Let's take a deeper dive into how integrating EHR interactions with expanded CMS API access can transform the developer landscape and, ultimately, beneficiary care. The goal isn't just to facilitate bit-level connectivity, it's about creating a robust ecosystem where data, technology, clinical insights, and patient outcomes interweave to deliver substantial improvements in healthcare delivery.

1. Breaking Down Silos with Rich, Longitudinal Data
In-depth Data Integration:
Imagine an API environment where developers can combine granular Medicare claims data with detailed clinical records from EHRs. This isn't just a matter of aggregating records; it's about creating a longitudinal health narrative for each beneficiary. With deep datasets, developers can craft algorithms that detect subtle changes in patient conditions, support chronic disease management, and forecast potential complications well before they escalate. Developers can harness machine learning algorithms to uncover hidden patterns that would otherwise remain obscured by data silos. This level of integration offers a unique value proposition, elevating the potential for precision medicine initiatives and personalized treatment plans.

Implications for Developers:
- **Advanced Analytics:** Access to both structured (e.g., claims, billing) and unstructured data (e.g., clinical notes) opens the door for sophisticated predictive analytics and natural language processing techniques.

- **Holistic Solutions:** Developers can design apps that not only pinpoint issues early but also suggest tailored interventions, ensuring that alerts, recommendations, or care plan adjustments consider the full spectrum of patient data.
- **Research and Innovation:** Rich datasets stimulate academic and private research, leading to breakthroughs that can be quickly translated into practical, deployable tools.

2. Redefining Developer Engagement with Interoperability Standards
Interoperability as a Catalyst:
Standardization through FHIR and other interoperability frameworks is essential for scaling digital health solutions. By enforcing strict, common data exchange protocols across EHRs, CMS can minimize the integration friction that typically discourages developers. When APIs adhere to universal standards, developers waste less time on interfacing challenges and more on building innovative, clinically grounded applications.

Developer Benefits:
- **Consistency and Predictability:** Developing against a reproducible interface means that solutions can be deployed across multiple systems without painstaking reconfiguration.
- **Rapid Prototyping and Iteration:** With well-documented, standard-compliant APIs, developers can iterate quickly, test hypotheses, and refine their applications with confidence.
- **Community Collaboration:** A standard environment fosters a collaborative ecosystem where shared tools, libraries, and frameworks emerge, further fueling the pace of innovation.

3. Emphasizing Patient-Centric Data Control and Privacy
Empowering Patients While Engaging Developers:
A deeper integration of EHR data demands a robust model for patient authorization and privacy. For developers, the opportunity lies in building utility-rich tools that respect and enhance patient agency. Consider patient-controlled data hubs where beneficiaries can grant, revoke, or modify access to their health records. This dynamic not only reinforces compliance with HIPAA and evolving privacy laws but also strengthens trust between patients, healthcare providers, and digital solution providers.

What This Means for Developers:
- **Trust as Currency.** Tools that transparently manage data permissions can become the gold standard, setting new benchmarks for privacy-focused design in healthcare.
- **Innovative Consent Models.** Building innovative consent frameworks into apps—perhaps using blockchain verifiable logs—can offer auditable patient data-sharing, opening new avenues for secure data exchange.
- **Value-Added Services.** Integrating patient-reported outcomes seamlessly with clinical data provides a richer picture of treatment efficacy, paving the way for apps

that not only monitor but actively involve beneficiaries in decision-making processes.

4. Fostering a Sustainable Ecosystem for Innovation
Long-Term Ecosystem Growth:
A truly deep integration is not a one-off achievement—it's a step toward a continually evolving ecosystem. CMS, by pairing its data with EHR interaction capabilities, can act as both an enabler and quality gatekeeper. In the long term, this integrated approach could evolve into a dedicated app marketplace or certification program where digital health solutions undergo rigorous evaluation to ensure they meet clinical, security, and efficacy standards.

Ecosystem Benefits to Developers:

- **Credibility and Trust.** Certification and quality marks instill confidence in patients and providers, helping new tools gain traction more swiftly.
- **Collaborative Innovation.** A defined ecosystem encourages partnerships between established tech companies, health systems, and emerging startups.
- **Feedback Loops and Continual Improvement.** Through real-world use, integrated data can generate feedback loops that allow for continual software and AI model refinement, keeping solutions current and clinically relevant.

Integrating these deep layers of EHR interaction with CMS data doesn't just serve as an upgrade—it reimagines the value chain of digital health. It transforms isolated data points into a symphony of insights, empowering developers to build tools that not only save costs and enhance care but also pioneer a new standard for the future of Medicare.

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

When we think about supplementing the Blue Button 2.0 API, it's not just about adding extra bits of data about layering context, nuance, and actionable information that can transform raw records into insights that drive better healthcare outcomes. Here are some of the most valuable complementary data sources:

1. Electronic Health Records (EHR) Data

What It Adds: EHRs provide detailed clinical information that goes far beyond the billing and claims data available via Blue Button 2.0. This includes lab results, vital signs, medications administered during visits, imaging reports, and even providers' clinical notes. When integrated with claims data, developers gain a holistic view of a beneficiary's health trajectory.
Why it's Valuable:

- **Rich Clinical Context.** Helps construct a comprehensive patient narrative that aids in early detection of health issues and supports personalized interventions.

- **Enhanced Predictive Analytics.** Merging clinical data with historical claims data gives developers the chance to build more robust risk assessment and decision-support tools.

2. Social Determinants of Health (SDOH) Data

What it Adds: SDOH data encompasses factors like socioeconomic status, housing stability, neighborhood environment, food security, and access to transportation. This data might come from government databases, community health surveys, or even credit and housing records aggregated and anonymized for health research.
Why it's Valuable:

- **Context and Holistic Understanding.** Provides the context necessary to understand why certain health outcomes occur, enabling digital solutions to tailor interventions based on an individual's environmental and social context.
- **Targeted Interventions.** Helps develop tools that can predict risk or recommend community-level resources, greatly enhancing the scope of personalized care.

3. Patient-Reported Outcomes (PROs) and Patient-Reported Experience Measures (PREMs)

What it Adds: These datasets capture information directly from beneficiaries about their symptoms, quality of life, and satisfaction with care. They can be gathered through surveys, mobile health applications, or wearable devices integrated into health monitoring platforms.
Why it's Valuable:

- **Direct Feedback Loop.** Offers insights into the effectiveness of treatments from a patient's perspective, which is often missing from claims data.
- **Improved Care Adjustment.** Enables developers to build adaptive systems that adjust care recommendations based on ongoing, real-time feedback from the patients themselves.

4. Provider Quality and Performance Metrics

What it Adds: Data on provider performance, including star ratings, patient satisfaction scores, adherence to clinical guidelines, and outcomes data, provides a layer of quality assurance. This information might come from CMS quality reporting programs, accreditation bodies, or independent quality assurance audits.
Why it's Valuable

- **Informed Decision-Making.** Enables developers to create tools that help beneficiaries easily identify high-performing providers, supporting better-informed choices in healthcare.
- **Benchmarking and Incentivization.** Serves as a foundation for creating value-based care dashboards or tools that reward quality improvement initiatives.

5. Cost Transparency and Reimbursement Data

What it Adds: Detailed insights into the costs of procedures, local reimbursement rates, and regional variations in pricing can be a game-changer. Such data might originate from state health departments, voluntary price reporting initiatives, or third-party aggregators.

Why it's Valuable

- **Financial Planning and Transparency.** Assists beneficiaries and providers in anticipating out-of-pocket expenses, ensuring cost-effective care decisions.
- **Policy Innovation.** Fuels the development of applications that monitor and compare healthcare costs, driving initiatives toward more equitable and affordable Medicare services.

6. Public Health and Population Health Data

What it Adds: Epidemiological data including disease prevalence, vaccination rates, and public health surveillance data can offer broader community-level insights. This type of information is typically sourced from organizations like the CDC, state health departments, and local public health agencies.

Why it's Valuable

- **Broader Context.** Enhances developers' ability to incorporate population health trends into their algorithms, making predictive models more accurate and reflective of real-world conditions.
- **Preemptive Intervention.** Supports the development of digital tools that can alert healthcare providers and patients about emerging health threats or seasonal patterns.

By creating a **data ecosystem** where these varied sources integrate seamlessly with Blue Button 2.0, CMS could empower developers to build applications that do more than just present data. Solutions could instead offer precision care, tailoring interventions, improving provider decision-making, and ultimately ensuring that Medicare beneficiaries receive the most effective, personalized care possible.

c. What obstacles prevent accessing these data sources today?

Accessing the richest and most actionable datasets alongside Blue Button 2.0 isn't simply a matter of making them available; several systemic, technical, and regulatory obstacles stand in the way:

1. **Regulatory and Privacy Concerns**
   - **HIPAA and Data Security:** Strict privacy laws such as HIPAA necessitate rigorous safeguards to ensure beneficiary data is anonymized and secure. The fear of data breaches and misuse imposes legal and administrative hurdles, causing many organizations to be reluctant in sharing detailed patient data without robust guarantees.
   - **Consent and Patient Authorization**: Building systems that allow for dynamic, patient-driven consent is complex. Determining what data beneficiaries are comfortable sharing—and ensuring this remains compliant with evolving legal standards—is an ongoing challenge for any comprehensive data integration effort.

2. **Data Fragmentation and Standardization Issues**

- **Multiplicity of Data Formats**: Data sources like EHRs, SDOH indicators, and public health records often stem from different vendors or entities and follow inconsistent standards, making unified access and aggregation difficult. While FHIR is a step in the right direction, legacy systems and non-standardized formats still erect significant barriers.
- **Siloed Data Systems**: Data across clinical, claims, and community platforms exist in silos. Lacking a centralized or coordinated approach, these systems often don't communicate well with one another. Developers are forced to build custom interfaces and mappings, significantly increasing both complexity and cost.

3. Interoperability and Technical Integration

- **Inconsistent API Implementation**: Even when APIs exist, the lack of uniformity across systems limits how seamlessly data can be shared. Variations in data quality, update frequency, and API response formats can mean that developers must account for numerous exceptions, adding layers of complexity to what ideally be straightforward data interactions.
- **Legacy System Limitations**: Many healthcare systems still rely on older, bespoke solutions that aren't built for modern, scalable API interactions. Integrating these legacy sources with newer, standardized datasets requires additional layers of middleware and data transformation.

4. Governance and Stakeholder Resistance

- **Data Ownership and Control:** With multiple stakeholders involved, from federal agencies to private health systems, deciding who controls the data and sets the terms of access can lead to prolonged negotiations. Concerns over data monetization, intellectual property rights, and competitive advantage among private vendors and healthcare providers further complicate collaboration.
- **Inadequate Incentives:** Without strong policy incentives or clear benefits that outweigh the costs and risks of sharing, various entities may prefer maintaining data silos rather than contributing to a universally accessible ecosystem. This lack of alignment in goals between CMS, providers, and third-party developers can slow progress.

These obstacles collectively highlight the significant effort required to create a fully integrated, interoperable data ecosystem. Overcoming them requires policy innovation, investment in technical infrastructure, and a commitment to collaborative, cross-sector transparency.

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

Expanding the API ecosystem under CMS and ASTP/ONC program policies can lead to a transformative increase in innovation for both patients and providers. Beyond Blue Button 2.0, here are several high-impact API areas to consider:

1. Enhanced Clinical Interoperability APIs

- **EHR Integration and FHIR Expansion API**: Implement APIs that harness advanced FHIR capabilities, allowing seamless exchange of clinical data between disparate EHR

systems. This API would not only support sharing structured data (such as med lists, lab results, and imaging) but also unstructured data (like clinical notes) to drive richer clinical decision support and patient care.

- **Clinical Decision Support and Guideline API:** An API that aggregates clinical guidelines, evidence-based practices, and real-time decision support prompts from various bodies (including CMS, professional societies, and research institutions) can help providers make informed, timely decisions at the point of care.

2. Patient-Centric and Engagement APIs

- **Patient-Reported Outcome (PRO) and Experience API**: This API would collect patient feedback from digital health applications, wearable devices, and surveys. By integrating direct patient input into medical records, developers can create applications that evolve with patient-reported outcomes, leading to continual improvements in care quality and personalization.
- **Consent and Data Authorization API:** Empower patients further by developing dynamic consent management systems. APIs in this space would handle granular permission settings, allowing patients to control which parts of their data (clinical, claims, SDOH) are shared with providers and third-party applications—all the while ensuring HIPAA-compliance and regulatory oversight.

3. Cost Transparency and Financial APIs

- **Payment and Claims Reconciliation API**: Beyond just claims data, this API could integrate real-time reimbursement and cost data, bridging the gap between clinical care and financial accountability. Providers could quickly reconcile expenses against payouts, and patients would be better informed about potential out-of-pocket costs before incurring them.
- **Pharmacy Benefit and Medication Adherence API**: Streamlining interactions between pharmacies, insurers, and patients, this API would provide real-time data on drug pricing, formulary status, and medication adherence, supporting tools that guide beneficiaries in making safe and cost-effective medication choices.

4. Social Determinants of Health and Community Health APIs

- **SDOH Integration API**: Incorporate data on socioeconomic indicators, housing stability, food security, and transportation. This API would allow developers to design interventions that target not just clinical symptoms but the underlying social factors that influence health outcomes, facilitating a more holistic approach to patient care.
- **Population Health and Public Health Surveillance API**: Leveraging epidemiological data from sources such as the CDC and local public health agencies, this API could fuel region-specific dashboards and predictive models that allow providers and policymakers to respond to emerging health trends or outbreaks proactively.

5. Provider Performance and Quality Measurement APIs

- **Provider Quality and Outcome API**: An API offering up-to-date quality metrics, satisfaction ratings, and outcome measures would help patients choose reliable providers and simultaneously motivate practices to continuously improve care

standards. This data could be derived from CMS quality programs, accreditation bodies, or even patient survey results.

- **Telehealth and Remote Patient Monitoring (RPM) API**: With telemedicine and remote monitoring becoming integral to care, an API that connects digital health platforms to provider systems would enhance appointment scheduling, connectivity for virtual visits, and real-time RPM data integration. This ensures greater continuity of care regardless of physical location.

6. Longitudinal Data and Advanced Analytics APIs

- **Integrated Longitudinal Health Record API**: By combining historical claims data, clinical records, and patient-reported metrics into a continuous timeline, this API would empower advanced analytics and predictive modeling. Developers could build applications that identify risk factors over time and deliver personalized care recommendations based on cumulative insights.
- **Artificial Intelligence and Machine Learning API**L Offering access to de-identified, aggregated datasets alongside tools for training and deployment of AI models, this API could accelerate the development of predictive analytics applications. Such tools can forecast health risks, patient readmission probabilities, or the efficacy of intervention strategies.

These additional APIs not only lower the barriers for developers by offering more granular, interoperable, and actionable data, but they also empower patients and providers to make decisions based on a full spectrum of information. The true value here lies in transforming data from a static resource into dynamic, interconnected streams that drive better care and outcomes.

3. Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

Let's zero in on publicly available APIs and how CMS can turbocharge their adoption to nurture innovation. Focusing on these APIs builds a level playing field for all developers and encourages the creation of interoperable, patient-centric solutions. Here's how CMS can make publicly available, standards-based APIs a cornerstone of the future healthcare data ecosystem:

1. Establish Robust, Transparent, and Developer-Friendly API Portals

- **Centralized API Hubs**: Create a comprehensive, easy-to-navigate developer portal dedicated to publicly available APIs. This portal should host detailed documentation, SDKs, sample code, and interactive sandbox environments that emulate real-world EHR and CMS interactions. Making it clear that the APIs are open and free encourages diverse development, from established companies to startups.
- **Versioning and Change Logs**: Transparently document updates, deprecations, and improvements so that developers can confidently integrate these APIs without worrying about unexpected changes.

2. Strengthen Open Standards and Interoperability

- **FHIR and Beyond:** Build on existing standards such as FHIR to ensure that publicly available APIs for EHR interactions are robust, consistent, and secure. By strictly adhering to open interoperability standards, CMS ensures that all provided APIs communicate seamlessly across systems and platforms.
- **Community-Driven Data Models:** Engage with standards organizations like HL7 and ONC to refine the data models used. A living standard maintained in collaboration with the developer community will facilitate broader adoption and evolution over time.

3. Financial and Regulatory Incentives

- **Funding and Rewards for Open API Projects:** Introduce grant programs, challenge competitions, and accelerated review processes that reward projects built exclusively on publicly available APIs. Financial support signals CMS's commitment and helps offset risks associated with investing in open, non-proprietary systems.
- **Linking Compliance with Value-Based Payments:** Tie open API usage into value-based reimbursement models. For example, demonstrate that efficiently integrated care systems using publicly available APIs can lead to better patient outcomes and lower costs, making a business case for their adoption.

4. Enhance Security, Privacy, and Patient Control

- **Open, Yet Secure:** Publicly available APIs must meet the highest security standards. Consider sandbox environments for developers to test their implementations without exposing real patient data. Emphasize that security protocols and privacy respect are baked into every layer of the API.
- **Granular Consent Frameworks:** Integrate patient-controlled consent mechanisms into the API design. This level of transparency increases trust and adoption since beneficiaries will see their data being handled in a way that respects their privacy preferences.

5. Education, Outreach, and Community Engagement

- **Webinars, Workshops, and Hackathons:** Host regular training sessions and interactive events focused solely on publicly available API usage. These sessions can highlight best practices, success stories, and innovative applications, serving as both a learning opportunity and a networking forum.
- **Developer Ecosystem and Open Innovation Labs:** Establish forums, discussion boards, and user groups that encourage knowledge sharing among developers. Showcase successful projects that leverage publicly available APIs, highlighting the advantages of interoperability and transparent data flows.

6. Streamline Procurement and Vendor Alignment

- **API-Centric Vendor Requirements:** When CMS or associated programs engage with vendors or technology partners, enforce standards that favor open, publicly available APIs. This creates a market advantage for solutions built on interoperable data and incentivizes vendors to align with these open practices.
- **Case Studies and Pilot Programs:** Launch and publish pilot projects that demonstrate the clear benefits of using publicly available APIs for EHR interactions. These real-

world examples can serve as a blueprint and a strong selling point for broader industry adoption.

By building a strong infrastructure around publicly available APIs, CMS not only supports innovators but also fosters a healthcare ecosystem rooted in transparency, security, and interoperability. Such an approach minimizes vendor lock-in, accelerates technology adoption, and, most importantly, empowers beneficiaries with better, more personalized care.

***TD-5.* How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?**

A nationwide provider directory of FHIR endpoints presents a tremendous opportunity to boost interoperability and data access across healthcare. CMS should be at the helm of this initiative, drawing on its established authority, public health mandate, and deep expertise in program administration to drive a standardized, accessible, and robust directory.

1. **Empowering CMS as a Trusted Leader**
   - **Regulatory Mandate and Authority**
     CMS, given its responsibility for Medicare and Medicaid programs, is uniquely positioned to set the rules for interoperability. By publishing a nationwide provider directory, CMS can ensure the directory is built on principles that support improved patient outcomes and cost-effective care. Its regulatory authority allows it to mandate participation and compliance from healthcare providers and vendors, ensuring consistent adherence to open standards.
   - **Central Coordination and Standardization**
     CMS's involvement guarantees that the directory reflects federal policy and public health priorities. Central oversight by CMS would harmonize disparate FHIR implementations and drive the adoption of uniform data standards. This centralization would benefit all stakeholders by reducing variability and streamlining the connectivity of clinical systems nationwide.
   - **Public Trust and Transparency**
     The credibility of CMS as a governmental entity helps establish trust among patients, providers, and payers. When CMS publishes the directory, stakeholders can be confident that the information is unbiased, standardized, and maintained in the public's interest. Transparent governance under CMS ensures continuous oversight and accountability.

2. **Enhancing Access to Health Information**
   - **For Patients**
     A CMS-led directory simplifies the process of locating and accessing one's health records through interoperable systems. Patients gain direct insight into which providers hold valid FHIR endpoints, empowering them to connect with their healthcare data seamlessly and securely. This ease-of-access will lead to improved self-management of chronic conditions and more informed health decisions.

- **For Providers**
  Providers benefit from knowing exactly where to retrieve comprehensive clinical data. With a centralized CMS directory, clinicians can easily locate the correct FHIR endpoints to gather critical patient information from various systems. This unified approach shortens information gaps during transitions of care, reduces duplication of efforts, and improves overall care coordination.
- **For Payers**
  Payers can leverage real-time, standardized clinical data to validate treatment claims and better manage reimbursement processes. The CMS directory acts as a vital tool that bridges the gap between clinical documentation and claims processing, reinforcing value-based care models and reducing administrative overhead.

3. Publication and Cost Considerations

- **CMS as the Publisher**
  Given its existing expertise, regulatory clout, and commitment to public health outcomes, CMS is the ideal organization to publish and maintain the nationwide provider directory. Early collaboration with the Office of the National Coordinator for Health Information Technology (ONC) and other standards bodies such as HL7 would enhance the directory's technical soundness and community acceptance, but overall ownership should rest with CMS.
- **Ensuring Free Access**
  To maximize public benefit and promote the widespread use of interoperable health data, the directory should be provided free of charge. Representation as a public good strengthens CMS's role as an enabler of innovation rather than a gatekeeper of information. Funding for this directory could come from federal appropriations or public-private partnerships, ensuring no additional cost is passed on to patients, providers, or payers.

4. Implementation Strategies Led by CMS

- **API Portal and Sandbox Environments**
  CMS can create a centralized API portal that functions as the interface to the directory. This portal would not only list all FHIR endpoints but also provide robust documentation, testing sandboxes, and developer outreach programs. By hosting workshops and hackathons, CMS can fuel innovation while providing hands-on technical support to developers.
- **Monitoring and Quality Governance**
  With CMS at the helm, the nationwide directory can incorporate periodic audits and quality reviews to ensure data integrity and endpoint reliability. Leveraging CMS's data governance frameworks ensures that patient privacy is maintained and that the information remains current and accurate.
- **Pilot Programs and Expansion Pathways**
  Prior to nationwide deployment, CMS might launch targeted pilot programs in select regions. These pilots would test the model, incorporate stakeholder feedback, and demonstrate measurable advancements in care coordination that could serve as justification for broader implementation.

By centering this initiative with CMS, the directory becomes more than a list of endpoints—it evolves into a transformational tool for interconnecting the diverse components of the U.S. healthcare ecosystem. CMS's leadership will help overcome current fragmentation, fostering a stronger, more integrated environment where patients, providers, and payers all benefit from ready access to comprehensive, real-time health information.

TD-6. What unique interoperability functions does TEFCA perform?

a. What existing alternatives should be considered?

TEFCA, the Trusted Exchange Framework and Common Agreement, is designed as a national interoperability backbone that brings together disparate health information systems under a unified set of principles, governance, and technical standards. Here's how TEFCA stands out and what alternatives you might consider:

Unique Interoperability Functions of TEFCA

1. **Common Trust Framework**
   TEFCA establishes a baseline of trust among participants through common principles and standardized terms and conditions. This framework mandates the essential elements of openness, privacy, and security across all connecting networks, ensuring that each participant is held to uniform standards.
2. **Nationwide Connectivity**
   Rather than encouraging isolated or regional networks, TEFCA's scope is national. It provides a universal policy layer that works across various health information networks (HINs), effectively bridging the gaps among regional health information exchanges (HIEs), private networks, and individual provider systems. This nationwide approach facilitates smoother query-based and directed exchange of electronic health information.
3. **Interoperability Standards Enforcement**
   TEFCA is technology-neutral but strongly encourages the use of modern data exchange standards such as FHIR (Fast Healthcare Interoperability Resources) and the inclusion of data elements defined in USCDI (United States Core Data for Interoperability). This drive for standardization helps overcome the challenges posed by legacy systems and proprietary interoperability models.
4. **Data Blocking and Information Governance**
   One of TEFCA's key roles is to minimize information blocking by establishing penalties and compliance guidelines. It creates mechanisms that encourage, and even require, the sharing of data across different health systems while still protecting patient privacy. This aspect is critical in fostering an environment where health information flows freely yet securely.
5. **Support for Both Query-Based and Directed Exchange**
   TEFCA accommodates multiple exchange methods. It supports on-demand query-based exchanges (where users can retrieve data from a network on-demand) alongside push-based directed messaging. This flexibility ensures the system can serve a broad range of use cases from care coordination to public health reporting and patient access to their own data.

Existing Alternatives to TEFCA

While TEFCA is crafted to serve as a comprehensive national solution, several existing alternatives and complementary models help move interoperability forward:

1. **Regional Health Information Exchanges (HIEs)**
   Many states or regions already have mature HIEs that provide localized interoperability. These networks are designed to address the unique needs of their communities. Although effective regionally, they often face challenges when scaling national connectivity, a gap that TEFCA seeks to fill.
2. **CommonWell Health Alliance**
   A private-sector collaboration, CommonWell focuses on linking health data across different provider networks. Accredited for its interoperability achievements, it allows for seamless data sharing between participating organizations but tends to be more vendor-specific compared with the open, standards-based approach of TEFCA.
3. **Carequality**
   Carequality is another consortium that has built a trust framework and common interoperability agreements to enable data sharing across different systems and vendors. Like CommonWell, it operates in the private sector but can be seen as a complement to TEFCA in the broader interoperability ecosystem.
4. **Proprietary Systems (e.g., Epic's Care Everywhere)**
   Many large vendors have developed their own proprietary networks to facilitate interoperability among their customers. While these systems often deliver robust performance within a closed ecosystem, they tend to limit access to outside systems and reduce cross-platform compatibility.
5. **Direct Messaging Networks**
   Though more limited in scope, direct messaging protocols have long been used for secure email-like communication between providers. They offer a basic level of interoperability, primarily for secure document exchange, but lack the comprehensive, nationwide scope that TEFCA is designed to provide.

Integrating TEFCA with Existing Alternatives

While TEFCA isn't necessarily meant to replace all existing alternatives overnight, it can serve as a unifying backbone that connects these various networks:

- Leverage Mature Regional Solutions: While regional HIEs continue to operate effectively within their spheres, TEFCA can integrate these systems into a cohesive national exchange.
- Bridge Proprietary Networks: Initiatives like CommonWell and Carequality have already created trusted frameworks for connectivity; aligning them with TEFCA's standards would amplify nationwide interoperability.
- Harmonize Diverse Exchange Models: By setting common principles, TEFCA helps ensure that query-based exchanges, direct exchange methods, and even traditional direct messaging systems can interoperate under a unified mission, enhancing overall data accessibility.

By performing these unique functions, TEFCA plays a pivotal role in reducing fragmentation in the health IT ecosystem and ensuring health information is accessible, secure, and usable across the entire nation. At the same time, recognizing and integrating with existing alternatives—each with its own strengths—creates a more resilient and adaptable infrastructure.

b. Are there redundant standards, protocols or channels or both that should be consolidated?

There are several areas where redundant standards, protocols, and channels have emerged over the years. Consolidating these can simplify the healthcare data exchange landscape and reduce integration burdens. This includes:

1. Messaging and Data Exchange Standards

- **HL7 v2 versus HL7 CDA versus FHIR**
  - HL7 v2 has been the workhorse for decades, especially for real-time transactional messages such as patient admissions, discharges, and transfers (ADT messages).
  - Clinical Document Architecture (CDA) was developed for more structured document exchange, such as discharge summaries and progress notes.
  - FHIR (Fast Healthcare Interoperability Resources), on the other hand, offers a modern, flexible approach that can encapsulate both transactional data and document exchange within one unified resource model.

    Specifics: As many systems still run on HL7 v2 and CDA, pushing for FHIR adoption—with well-defined USCDI (United States Core Data for Interoperability) profiles—could consolidate these older standards into a single, more agile framework.
- **Direct Messaging versus Query-Based Exchange**
  - Direct Secure Messaging is a secure email-like protocol used for one-to-one communication, often seen as a stopgap solution.
  - FHIR Query APIs and TEFCA-based exchanges are designed to support real-time, context-aware queries that can access patient data when needed.

    Specifics: Instead of maintaining two different channels (one for bulk secure sending and one for on-demand queries), aligning capabilities under TEFCA's broader framework could reduce overlapping functionality.

2. Interoperability Channels and Networks

- **Proprietary Networks versus National Frameworks**
  - Vendor-Specific Networks: Systems like Epic's Care Everywhere, Cerner's exchange networks, and even independent private networks like those facilitated by the CommonWell Health Alliance or Carequality are often limited to participants within a closed circle.
  - Nationwide Backbone (TEFCA): TEFCA aims to provide a single, uniform framework that can bridge regional and vendor-specific HIEs on a national

level.

Specifics: Consolidating these channels under a nationwide framework would improve connectivity. For example, rather than multiple overlapping agreements and proprietary solutions, stakeholders could align with TEFCA's common trust framework and technical standards, reducing redundancy while still preserving the option for specialized networks in niche areas.

3. Clinical Content and Documentation Frameworks

- **Overlapping Document Standards:**
  - Multiple structured formats such as CDA documents, DICOM (for images), and narrative documents from different systems often represent the same clinical concepts in different ways.
  - FHIR Resources can support a wide range of clinical content, from laboratory results and medication lists to full clinical notes.

    Specifics: By promoting FHIR's resource-based approach, CMS and other governing bodies can move toward a unified method for clinical documentation, thereby simplifying the standardization process across various types of clinical data.

4. Governance and Policy Consolidation

- **Fragmented Regulatory Guidance**
  - Multiple governmental and industry efforts sometimes offer overlapping or even conflicting guidance, which creates both technical and administrative burdens on healthcare providers and vendors.
  - Unified Oversight from CMS and ONC: If CMS, along with the Office of the National Coordinator (ONC), spearheads the consolidation, it could mandate the use of open, FHIR-based, TEFCA-aligned solutions and provide clear, centralized guidelines on standards adoption.

    Specifics: An authoritative push for standardized APIs and interoperability specifications—backed by funding incentives and regulatory mandates—could gradually phase out redundant practices and force a more streamlined, interoperable ecosystem.

Conclusion. Consolidating overlapping standards, such as merging HL7 v2 and CDA functions into a comprehensive FHIR-based approach, and unifying disparate channels (like direct messaging versus TEFCA-driven exchanges) can lead to:

- Reduced complexity for developers and healthcare organizations (fewer interfaces to integrate and maintain).
- Enhanced patient safety and data reliability (consistent interpretation of clinical data).
- Greater nationwide interoperability (a truly connected healthcare system where data flows efficiently and securely).

By undertaking these consolidation efforts under the leadership of CMS, supported by clear regulatory frameworks and collaboration with industry consortia such as HL7, CommonWell, and Carequality, the healthcare ecosystem can move toward a more efficient, robust data exchange infrastructure.

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

The adoption of the U.S. Core Data for Interoperability (USCDI) has meaningfully advanced interoperability by establishing a standardized, baseline set of data classes that facilitate more consistent and structured information exchange across health IT systems. It has been especially effective in enhancing clinical document exchange, supporting care transitions, and enabling regulatory reporting.

However, several limitations remain that impact its utility in operational settings:

1. **Gaps in Granular, Context-Rich Data.** While USCDI includes core clinical elements, it lacks sufficient granularity for certain use cases, such as post-acute care planning, social determinants of health (SDOH), and functional status assessments, which are critical for whole-person care, particularly in value-based payment models.
2. **Specialty-Specific Needs Not Met.** USCDI is still relatively limited in supporting behavioral health, rehabilitation, and long-term services and supports (LTSS) data needs. For example, IRFs and SNFs rely on nuanced assessments and outcome measures (e.g., mobility scores, cognitive patterns) that are not fully captured in current USCDI versions.
3. **Inconsistent Implementation Across Systems.** The mere presence of USCDI elements does not guarantee consistent capture or availability across vendor platforms. There is often variation in how elements are structured, populated, or transmitted, especially when pulled from disparate EHR modules (e.g., nursing vs. medical records).
4. **Limitations in Data Provenance and Timeliness.** USCDI does not yet address data provenance, quality, or timeliness, all of which are critical in high-acuity settings where decisions depend on real-time, source-verified data.

a. Does it contain the full extent of data elements you need?

USCDI provides a valuable foundation, but it does not currently include the full extent of data elements needed for comprehensive care coordination, regulatory compliance, and quality improvement. We recommend CMS consider:

- Accelerating the adoption of USCDI+ for specialized settings (e.g., PAC, behavioral health)
- Expanding support for patient-reported outcomes, nutrition, functional goals, and environmental risks
- Encouraging alignment between USCDI data elements and CMS quality measures, to reduce dual reporting burdens

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?

Definition-related limitations: Some data elements are either too broadly defined or lack sufficient specificity to support consistent implementation across care settings. For example, "problems" or "functional status" may be captured differently depending on the assessment tools in use or clinical specialty.

Utilization-related limitations: Even where elements are well defined, real-world utilization is inconsistent due to variation in EHR vendor implementation, clinical workflows, and organizational capacity. Some systems treat USCDI compliance as a checkbox rather than fully integrating structured capture and exchange capabilities into their user interface or API strategies.

The lack of enforcement mechanisms or incentives tied to data completeness and accuracy further dilutes the potential of USCDI as a meaningful interoperability standard.

c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?

Adding more data elements could indeed add significant value, especially in areas such as:

- Post-acute care situations, in which nuanced functional status, discharge planning, and caregiver support data are essential
- Behavioral health situations, in which context and historical trajectories are often more important than discrete values
- Social determinants of health (SDOH) and environmental exposures, which are critical to risk stratification and care coordination

However, expanding USCDI does raise scoping and implementation challenges:

- Burden on implementers: More elements can overwhelm EHR vendors and providers if not clearly prioritized or supported by strong use cases.
- Data quality risks: Without careful curation, new data classes may be inconsistently captured, leading to increased noise instead of improved signal.
- Governance complexity: The more expansive USCDI becomes, the more difficult it is to ensure cross-stakeholder alignment.

How to address these challenges:

- Tiered adoption: USCDI could support a tiered model (e.g., "Core", "Extended", "Specialty Modules") to allow flexible implementation based on setting or use case.
- Piloting and feedback loops: New data elements should be piloted in real-world clinical or administrative settings to assess feasibility before national rollout.
- Stronger alignment with CMS programs: When data elements are tied to value-based care models, reporting requirements, or incentives, adoption becomes more purposeful.

d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

There is a growing recognition that natural language processing (NLP) and foundation models can enhance the value of less structured clinical narratives. In many cases, unstructured notes contain richer clinical context than discrete data elements—particularly for complex, longitudinal, or social care situations. That said, the preference for less structured formats comes with important caveats:

- Receiver readiness: Many systems (especially smaller providers or community-based organizations) lack the technical or financial capacity to deploy advanced NLP pipelines.
- Transparency and accountability: Structured data supports traceability and auditability, while extracted insights from free-text formats may introduce interpretive variability.
- Bias and fairness: Language models can reproduce or amplify biases unless trained and validated rigorously.

In general, we recommend a hybrid approach:
Continue expanding structured data elements in USCDI where feasible, especially for high-value, cross-setting use cases. Encourage use of NLP and AI for unstructured data extraction, but in parallel—not as a replacement—until the accuracy, explainability, and interoperability of those tools meet healthcare-grade standards.

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